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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Capacitor

JEE Main Physics · Previous Year Questions with Answer Key

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Numerical	50 questions
MCQ (Single Correct Answer)	133 questions
Answer key	at the end

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Numerical 50 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

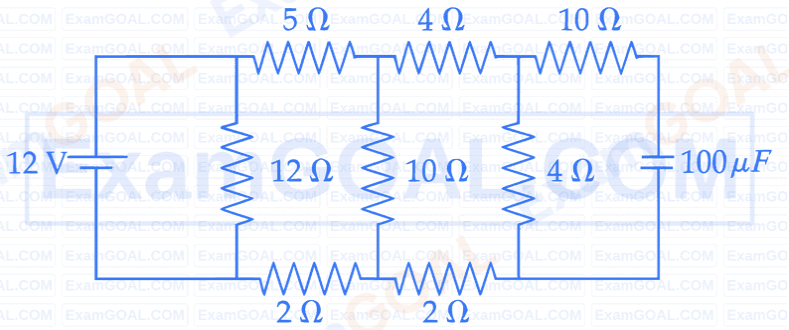
A parallel plate capacitor is having separation between plates 0.885 mm . It has a capacitance of $1\mu\,\mathrm{F}$ when the space between the plates is filled with an insulating material of resistivity $1 \times 10^{13}\Omega\,\mathrm{m}$ and resistance $17.7 \times 10^{14}\Omega$. Relative permittivity of the insulating material is $\alpha \times 10^{7}$. The value of α is ____ .

(Take permittivity of free space $= 8.85 \times 10^{-12}\,\mathrm{F/m}$)

Open solution

Q2. JEE Main 2026 (Online) 8th April Evening Shift

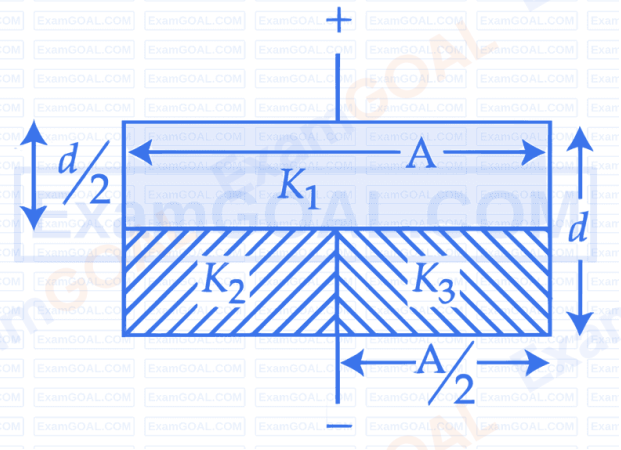
The stored charge in the capacitor in steady state of the following circuit is ____ $\mu\mathrm{C}$.



Open solution

Q3. JEE Main 2026 (Online) 23rd January Morning Shift

The space between the plates of a parallel plate capacitor of capacitance C (without any dielectric) is now filled with three dielectric slabs of dielectric constants $K_1 = 2, K_2 = 3$ and $K_3 = 5$ (as shown in figure). If new capacitance is $\frac{n}{3}C$ then the value of n is ____ .



Open solution

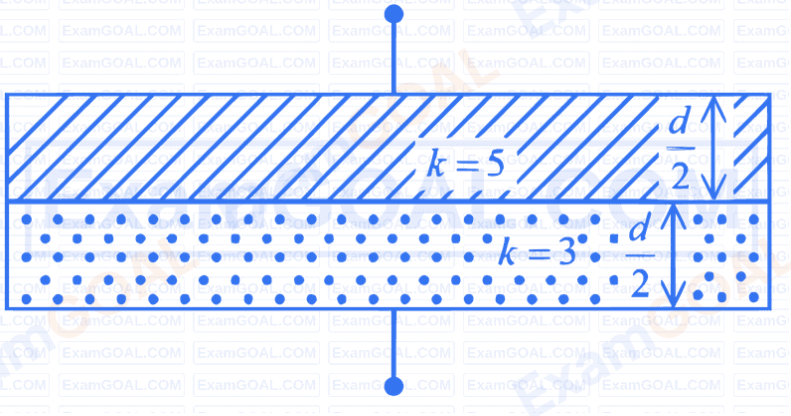
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Q4. JEE Main 2026 (Online) 22nd January Evening Shift

A capacitor P with capacitance 10×10^{-6} F is fully charged with a potential difference of 6.0 V and disconnected from the battery. The charged capacitor P is connected across another capacitor Q with capacitance 20×10^{-6} F. The charge on capacitor Q when equilibrium is established will be $\alpha \times 10^{-5} C$ (assume capacitor Q does not have any charge initially), the value of α is ____ .

Open solution

Q5. JEE Main 2025 (Online) 8th April Evening Shift



Space between the plates of a parallel plate capacitor of plate area 4 cm^2 and separation of 1.77 mm, is filled with uniform dielectric materials with dielectric constants (3 and 5) as shown in figure. Another capacitor of capacitance 7.5 pF is connected in parallel with it. The effective capacitance of this combination is _ pF.

(Given $\epsilon_0 = 8.85 \times 10^{-12}$ F/m)

Open solution

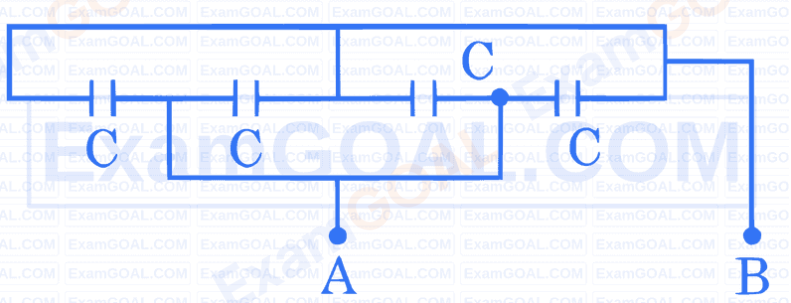
Q6. JEE Main 2025 (Online) 7th April Evening Shift

A parallel plate capacitor has charge 5×10^{-6}C. A dielectric slab is inserted between the plates and almost fills the space between the plates. If the induced charge on one face of the slab is 4×10^{-6}C then the dielectric constant of the slab is _____.

Open solution

Q7. JEE Main 2025 (Online) 4th April Morning Shift

Four capacitors each of capacitance $16\mu F$ are connected as shown in the figure. The capacitance between points A and B is : _____ (in μF).



Open solution

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Q8. JEE Main 2025 (Online) 29th January Evening Shift

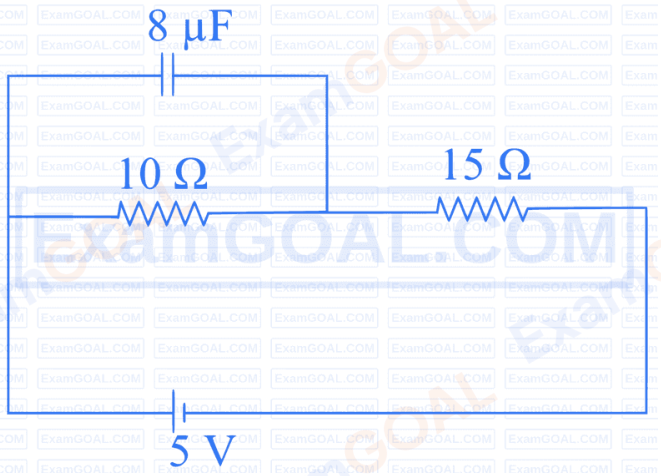
A parallel plate capacitor consisting of two circular plates of radius 10 cm is being charged by a constant current of 0.15 A . If the rate of change of potential difference between the plates is 7×10^8 V/s then the integer value of the distance between the parallel plates is

(Take, $\epsilon_0 = 9 \times 10^{-12} \frac{\text{F}}{\text{m}}, \pi = \frac{22}{7}$) _____ μm.

Open solution

Q9. JEE Main 2025 (Online) 23rd January Evening Shift

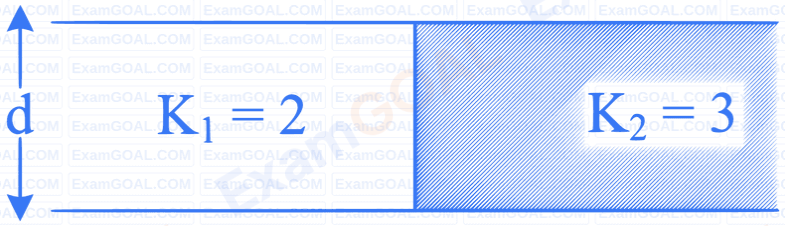
At steady state the charge on the capacitor, as shown in the circuit below, is _____ μC.



Open solution

Q10. JEE Main 2024 (Online) 6th April Evening Shift

A capacitor of 10μF capacitance whose plates are separated by 10 mm through air and each plate has area $4\ \text{cm}^2$ is now filled equally with two dielectric media of $K_1 = 2, K_2 = 3$ respectively as shown in figure. If new force between the plates is 8 N. The supply voltage is _____ V.



Open solution

Q11. JEE Main 2024 (Online) 5th April Morning Shift

The electric field between the two parallel plates of a capacitor of 1.5μF capacitance drops to one third of its initial value in 6.6μs when the plates are connected by a thin wire. The resistance of this wire is _____ Ω. (Given, $\log_e 3 = 1.1$)

Open solution

Q12. JEE Main 2024 (Online) 5th April Morning Shift

Three capacitors of capacitances 25μF, 30μF and 45μF are connected in parallel to a supply of 100 V. Energy stored in the above combination is E. When these capacitors are connected in series to the same supply, the stored energy is $\frac{9}{x}$E. The value of x is _____.

Open solution

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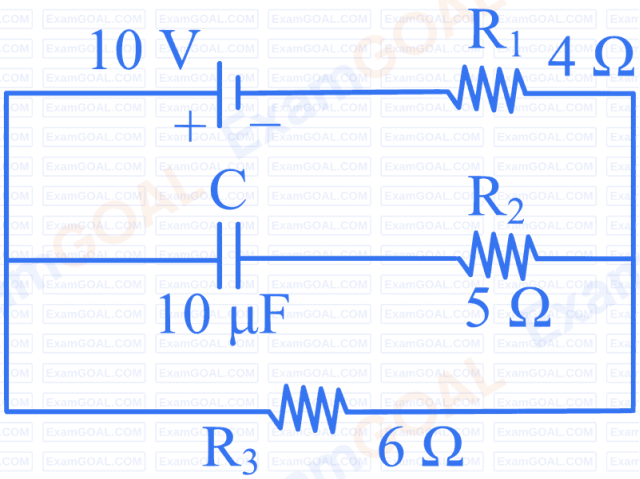
Q13. JEE Main 2024 (Online) 4th April Evening Shift

A parallel plate capacitor of capacitance $12.5\ \mathrm{pF}$ is charged by a battery connected between its plates to potential difference of $12.0\ \mathrm{V}$. The battery is now disconnected and a dielectric slab $(\epsilon_r = 6)$ is inserted between the plates. The change in its potential energy after inserting the dielectric slab is _____ $\times 10^{-12}\ \mathrm{J}$.

Open solution

Q14. JEE Main 2024 (Online) 1st February Evening Shift

In an electrical circuit drawn below the amount of charge stored in the capacitor is _____ μ C.



Open solution

Q15. JEE Main 2024 (Online) 31st January Morning Shift

A parallel plate capacitor with plate separation $5\ \mathrm{mm}$ is charged up by a battery. It is found that on introducing a dielectric sheet of thickness $2\ \mathrm{mm}$, while keeping the battery connections intact, the capacitor draws 25% more charge from the battery than before. The dielectric constant of the sheet is _____.

Open solution

Q16. JEE Main 2024 (Online) 30th January Morning Shift

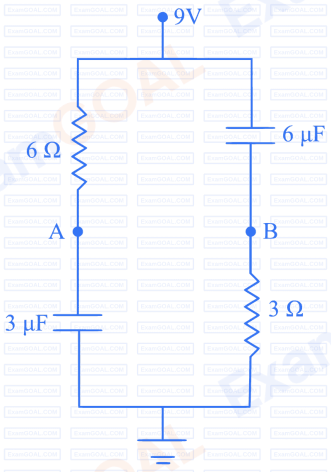
A capacitor of capacitance C and potential V has energy E. It is connected to another capacitor of capacitance $2\mathrm{C}$ and potential $2\ \mathrm{V}$. Then the loss of energy is $\frac{x}{3}\mathrm{E}$, where x is _____.

Open solution

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Q17. JEE Main 2024 (Online) 29th January Evening Shift

In the given figure, the charge stored in $6\mu F$ capacitor, when points A and B are joined by a connecting wire is _____ μC.



Open solution

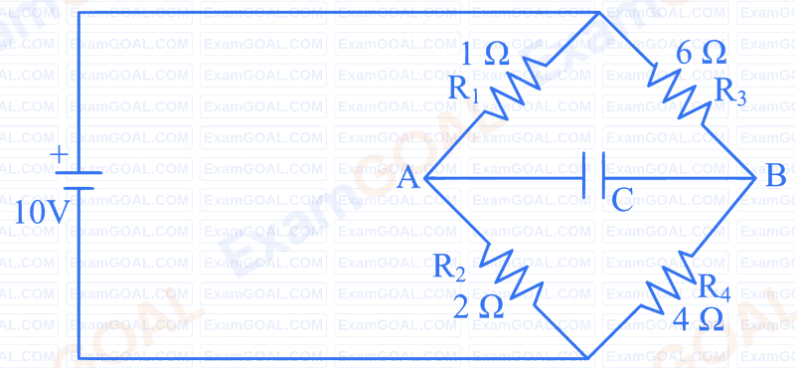
Q18. JEE Main 2024 (Online) 29th January Morning Shift

A 16Ω wire is bent to form a square loop. A 9 V battery with internal resistance 1Ω is connected across one of its sides. If a $4\mu F$ capacitor is connected across one of its diagonals, the energy stored by the capacitor will be $\frac{x}{2}\mu J$, where $x =$ _____

Open solution

Q19. JEE Main 2024 (Online) 27th January Morning Shift

The charge accumulated on the capacitor connected in the following circuit is _____ μC (Given $C = 150\mu F$)



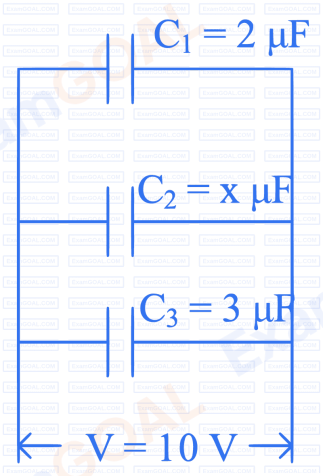
Open solution

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Q20. JEE Main 2023 (Online) 15th April Morning Shift

In the given figure the total charge stored in the combination of capacitors is $100\ \mu C$. The value of ' x ' is _____.

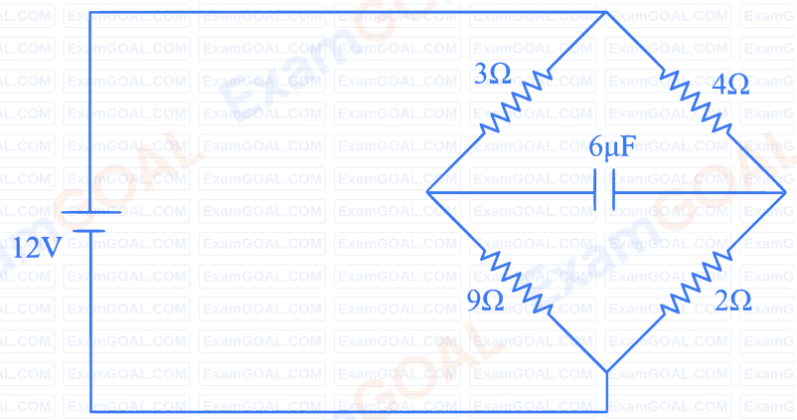
Open solution



Q21. JEE Main 2023 (Online) 13th April Evening Shift

In the circuit shown, the energy stored in the capacitor is $n\ \mu J$. The value of n is _____

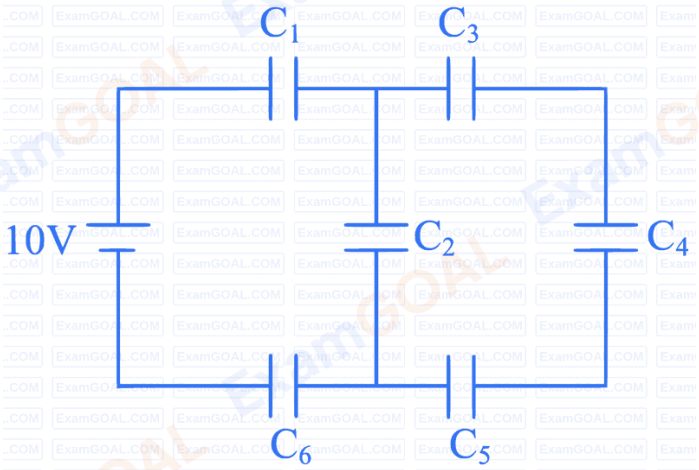
Open solution



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Q22. JEE Main 2023 (Online) 11th April Evening Shift

In the given circuit, $C_1 = 2\mu F, C_2 = 0.2\mu F, C_3 = 2\mu F, C_4 = 4\mu F, C_5 = 2\mu F, C_6 = 2\mu F$, The charge stored on capacitor C_4 is _____ μC.



Open solution

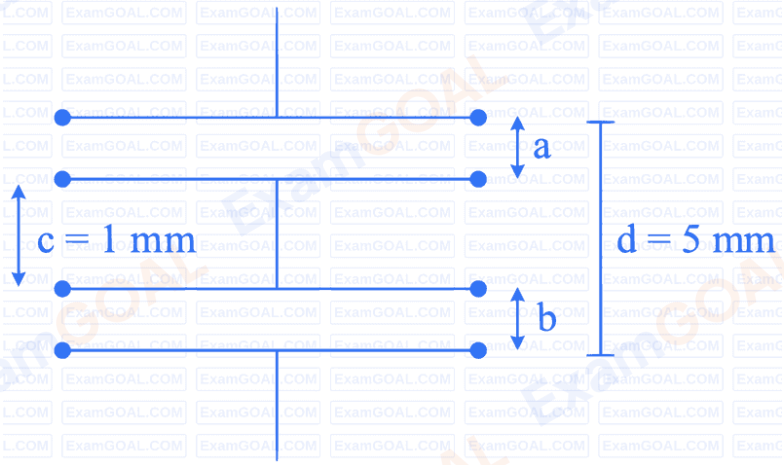
Q23. JEE Main 2023 (Online) 8th April Evening Shift

A $600\ \mathrm{pF}$ capacitor is charged by $200\ \mathrm{V}$ supply. It is then disconnected from the supply and is connected to another uncharged $600\ \mathrm{pF}$ capacitor. Electrostatic energy lost in the process is _____ μJ

Open solution

Q24. JEE Main 2023 (Online) 6th April Evening Shift

As shown in the figure, two parallel plate capacitors having equal plate area of $200\ \mathrm{cm}^2$ are joined in such a way that $a \neq b$. The equivalent capacitance of the combination is $x \in_0\ \mathrm{F}$. The value of x is _____.



Open solution

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Q25. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

A parallel plate capacitor with plate area A and plate separation d is filled with a dielectric material of dielectric constant $K = 4$. The thickness of the dielectric material is x, where $x < d$.

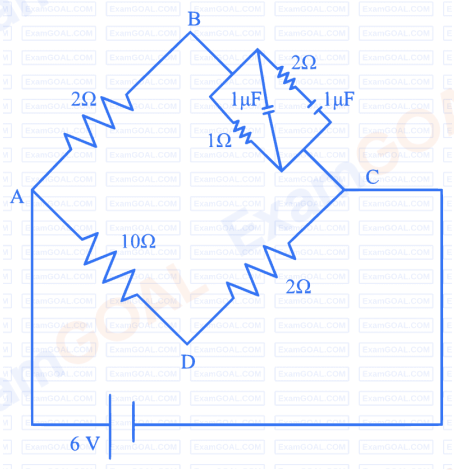


Let C_1 and C_2 be the capacitance of the system for $x = \frac{1}{3}d$ and $x = \frac{2d}{3}$, respectively. If $\mathrm{C}_1 = 2\mu\mathrm{F}$ the value of C_2 is _____ $\mu\mathrm{F}$

Q26. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

For the given circuit, in the steady state, $|\mathrm{V_B} - \mathrm{V_D}| =$ _____ V.



Q27. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

Two parallel plate capacitors C_1 and C_2 each having capacitance of $10\mu\mathrm{F}$ are individually charged by a 100 V D.C. source. Capacitor C_1 is kept connected to the source and a dielectric slab is inserted between its plates. Capacitor C_2 is disconnected from the source and then a dielectric slab is inserted in it. Afterwards the capacitor C_1 is also disconnected from the source and the two capacitors are finally connected in parallel combination. The common potential of the combination will be _____ V.

(Assuming Dielectric constant $= 10$)

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Q28. JEE Main 2023 (Online) 30th January Morning Shift

A capacitor of capacitance $900\mu F$ is charged by a $100\ V$ battery. The capacitor is disconnected from the battery and connected to another uncharged identical capacitor such that one plate of uncharged capacitor connected to positive plate and another plate of uncharged capacitor connected to negative plate of the charged capacitor. The loss of energy in this process is measured as $x \times 10^{-2}$ J. The value of x is _____.

Open solution

Q29. JEE Main 2023 (Online) 25th January Evening Shift

A capacitor has capacitance $5\mu F$ when its parallel plates are separated by air medium of thickness d. A slab of material of dielectric constant 1.5 having area equal to that of plates but thickness $\frac{d}{2}$ is inserted between the plates. Capacitance of the capacitor in the presence of slab will be _____ μF.

Open solution

Q30. JEE Main 2023 (Online) 24th January Evening Shift

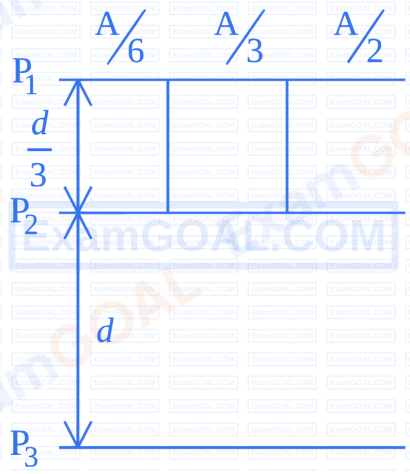
A parallel plate capacitor with air between the plates has a capacitance of 15pF. The separation between the plates becomes twice and the space between them is filled with a medium of dielectric constant 3.5. Then the capacitance becomes $\frac{x}{4}$ pF. The value of x is _____.

Open solution

Q31. JEE Main 2022 (Online) 30th July Evening Shift

Combination of four parallel plate air capacitors is shown in figure. Separation between the plate P_1 & P_2 is $\frac{d}{3}$ and separation between the plates P_2 and P_3 is d. The equivalent capacitance of the configuration is _____ μF.

(Given $\frac{\epsilon_0 A}{d} = 4\ \mu F$. Where A=Area of plates)

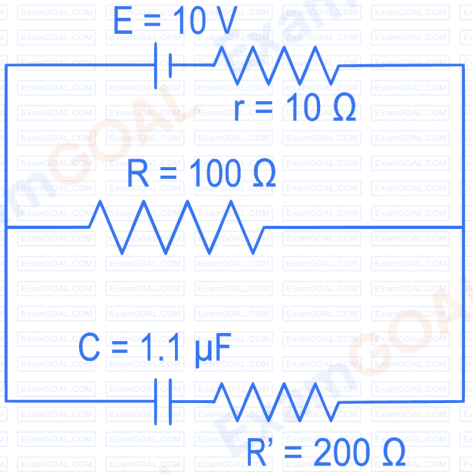


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Q32. JEE Main 2022 (Online) 27th July Evening Shift

As shown in the figure, in steady state, the charge stored in the capacitor is _____ $\times 10^{-6}$ C.



Open solution

Q33. JEE Main 2022 (Online) 27th July Evening Shift

A parallel plate capacitor with width $4\ \text{cm}$, length $8\ \text{cm}$ and separation between the plates of $4\ \text{mm}$ is connected to a battery of $20\ \text{V}$. A dielectric slab of dielectric constant 5 having length $1\ \text{cm}$, width $4\ \text{cm}$ and thickness $4\ \text{mm}$ is inserted between the plates of parallel plate capacitor. The electrostatic energy of this system will be _____ ϵ_0 J. (Where ϵ_0 is the permittivity of free space)

Open solution

Q34. JEE Main 2022 (Online) 26th July Morning Shift

A composite parallel plate capacitor is made up of two different dielectric materials with different thickness (t_1 and t_2) as shown in figure. The two different dielectric materials are separated by a conducting foil F. The voltage of the conducting foil is _____ V.



Open solution

Q35. JEE Main 2022 (Online) 25th July Evening Shift

Two parallel plate capacitors of capacity C and 3C are connected in parallel combination and charged to a potential difference 18 V. The battery is then disconnected and the space between the plates of the capacitor of capacity C is completely filled with a material of dielectric constant 9. The final potential difference across the combination of capacitors will be _____ V.

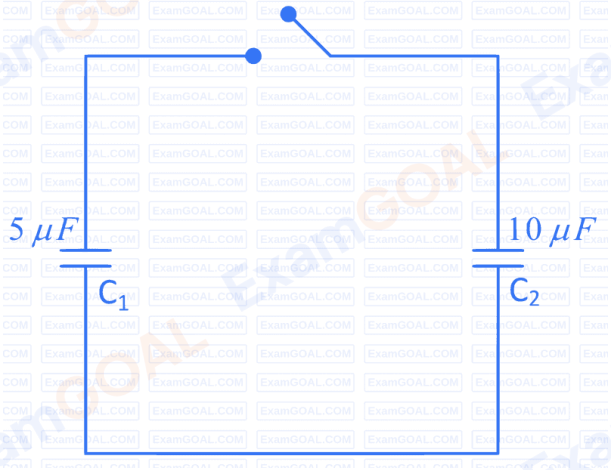
Open solution

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Q36. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

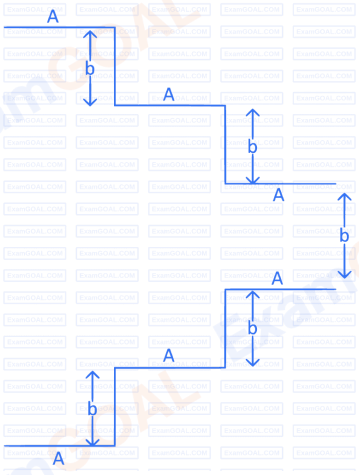
A capacitor C_1 of capacitance 5 μF is charged to a potential of 30 V using a battery. The battery is then removed and the charged capacitor is connected to an uncharged capacitor C_2 of capacitance 10 μF as shown in figure. When the switch is closed charge flows between the capacitors. At equilibrium, the charge on the capacitor C_2 is _____ μC.



Q37. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

A parallel plate capacitor is made up of stair like structure with a plate area A of each stair and that is connected with a wire of length b, as shown in the figure. The capacitance of the arrangement is $\frac{x}{15}\frac{\epsilon_0 A}{b}$. The value of x is _____.



Q38. JEE Main 2022 (Online) 27th June Morning Shift

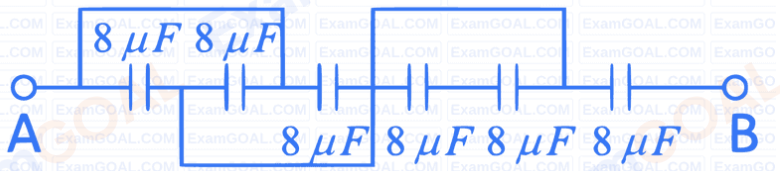
Open solution

A capacitor of capacitance 50 pF is charged by 100 V source. It is then connected to another uncharged identical capacitor. Electrostatic energy loss in the process is _____ nJ.

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Q39. JEE Main 2022 (Online) 25th June Morning Shift

The equivalent capacitance between points A and B in below shown figure will be _____ μF.



Open solution

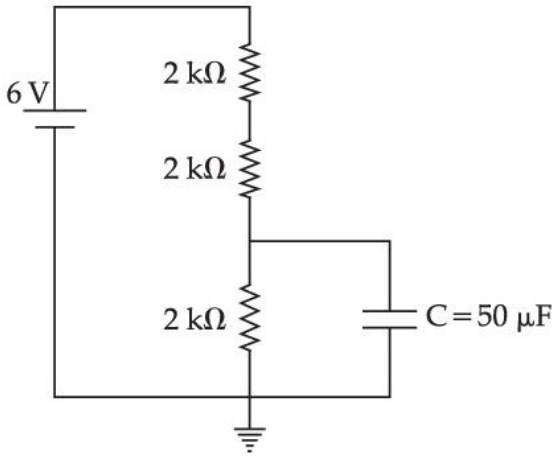
Q40. JEE Main 2021 (Online) 31st August Evening Shift

A parallel plate capacitor of capacitance 200 μF is connected to a battery of 200 V. A dielectric slab of dielectric constant 2 is now inserted into the space between plates of capacitor while the battery remain connected. The change in the electrostatic energy in the capacitor will be _____J.

Open solution

Q41. JEE Main 2021 (Online) 31st August Morning Shift

A capacitor of 50 μF is connected in a circuit as shown in figure. The charge on the upper plate of the capacitor is _____μC.



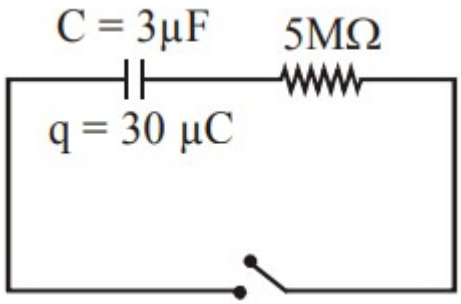
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Q42. JEE Main 2021 (Online) 18th March Morning Shift

The circuit shown in the figure consists of a charged capacitor of capacity 3 μF and a charge of 30 μC. At time t = 0, when the key is closed, the value of current flowing through the 5 MΩ resistor is 'x' μ-A. The value of 'x' to the nearest integer is _____.

Open solution



Q43. JEE Main 2021 (Online) 18th March Morning Shift

A parallel plate capacitor has plate area 100 m^2 and plate separation of 10 m. The space between the plates is filled up to a thickness 5 m with a material of dielectric constant of 10. The resultant capacitance of the system is 'x' pF.

The value of $\varepsilon_0 = 8.85 \times 10^{-12}$ F.m^{-1}.

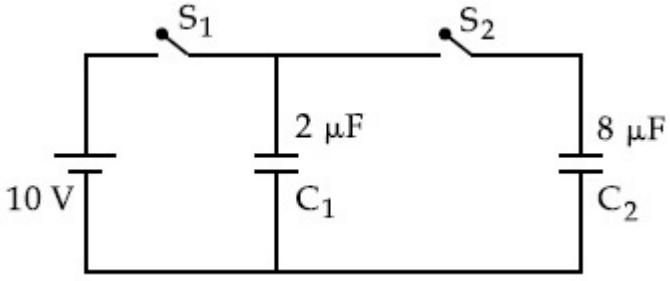
The value of 'x' to the nearest integer is _____.

Open solution

Q44. JEE Main 2021 (Online) 17th March Evening Shift

A 2μF capacitor C_1 is first charged to a potential difference of 10V using a battery. Then the battery is removed and the capacitor is connected to an uncharged capacitor C_2 of 8 μF. The charge in C_2 on equilibrium condition is _____ μC. (Round off to the Nearest Integer)

Open solution



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Q45. JEE Main 2021 (Online) 17th March Morning Shift

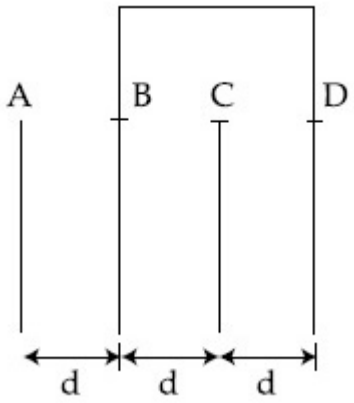
A parallel plate capacitor whose capacitance C is 14 pF is charged by a battery to a potential difference V = 12 V between its plates. The charging battery is now disconnected and a porcelain plate with k = 7 is inserted between the plates, then the plate would oscillate back and forth between the plates with a constant mechanical energy of _____ pJ. (Assume no friction)

Open solution

Q46. JEE Main 2021 (Online) 17th March Morning Shift

Four identical rectangular plates with length, l = 2 cm and breadth, b = $\frac{3}{2}$ cm are arranged as shown in figure. The equivalent capacitance between A and C is $\frac{x\varepsilon_0}{d}$. The value of x is _____. (Round off to the Nearest Integer)

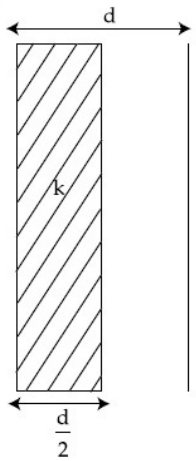
Open solution



Q47. JEE Main 2021 (Online) 16th March Evening Shift

In a parallel plate capacitor set up, the plate area of capacitor is 2 m^2 and the plates are separated by 1 m. If the space between the plates are filled with a dielectric material of thickness 0.5 m and area 2 m^2 (see fig.) the capacitance of the set-up will be _____ε_0. (Dielectric constant of the material = 3.2) (Round off to the Nearest Integer)

Open solution

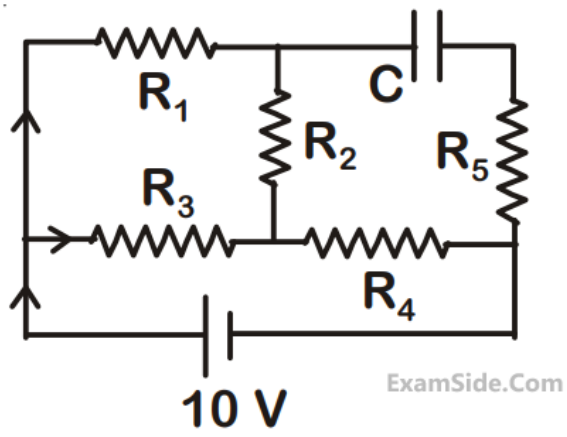


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Q48. JEE Main 2020 (Online) 2nd September Evening Slot

An ideal cell of emf 10 V is connected in circuit shown in figure. Each resistance is 2 Ω. The potential difference (in V) across the capacitor when it is fully charged is _____.

Open solution



Q49. JEE Main 2020 (Online) 2nd September Morning Slot

A 5 μF capacitor is charged fully by a 220 V supply. It is then disconnected from the supply and is connected in series to another uncharged 2.5 μF capacitor. If the energy change during the charge redistribution is $\frac{X}{100}J$ then value of X to the nearest integer is _____.

Open solution

Q50. JEE Main 2020 (Online) 7th January Evening Slot

A 60 pF capacitor is fully charged by a 20 V supply. It is then disconnected from the supply and is conneced to another uncharged 60 pF capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ) _____

Open solution

MCQ (Single Correct Answer) 133 questions

Q51. JEE Main 2026 (Online) 6th April Evening Shift

A sphere of capacitance 100 pF is charged to a potential of 100 V . Another identical uncharged metal sphere is brought in contact with the charged sphere, then the change in the total energy stored on these spheres, when they touch is $\alpha \times 10^{-7}$ J. The value of α is ____ .

(combined capacitance of spheres is 200 pF)

(A) 5 (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

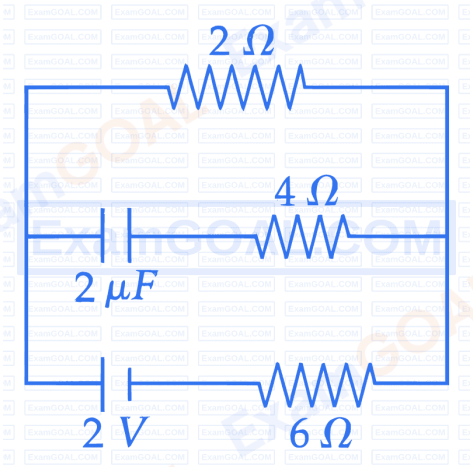
Open solution

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Q52. JEE Main 2026 (Online) 5th April Evening Shift

Under steady state condition the potential difference across the capacitor in the circuit is ____ V.

Open solution



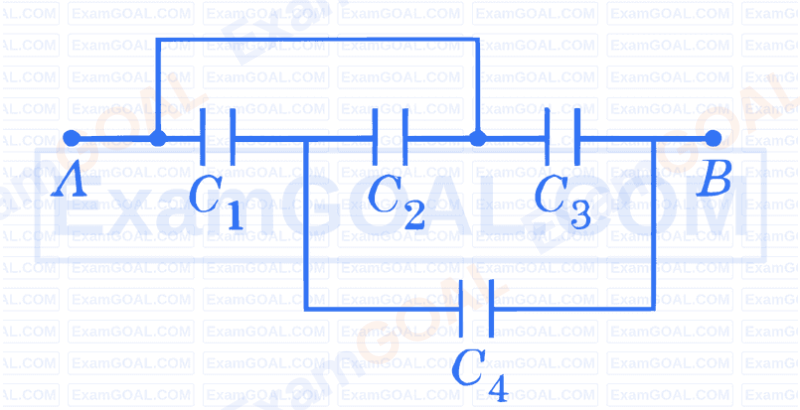
(A) 0.5 (B) 1.5 (C) 0 (D) 2

Q53. JEE Main 2026 (Online) 4th April Evening Shift

From the circuit given below, the capacitance between terminals A and B shown in the circuit is ____ μF.

(take $C_1 = C_2 = C_3 = 1\mu\,\mathrm{F}$ and $C_4 = 2\mu\,\mathrm{F}$.)

Open solution



(A) 2 (B) 7/2 (C) 7/3 (D) 5/2

Q54. JEE Main 2026 (Online) 4th April Morning Shift

A parallel plate air capacitor is connected to a battery. The plates are pulled apart at uniform speed v. If x is the separation between the plates at any instant, then the time rate of change of electrostatic energy of the capacitor is proportional to x^{α}, where α is ____ .

Open solution

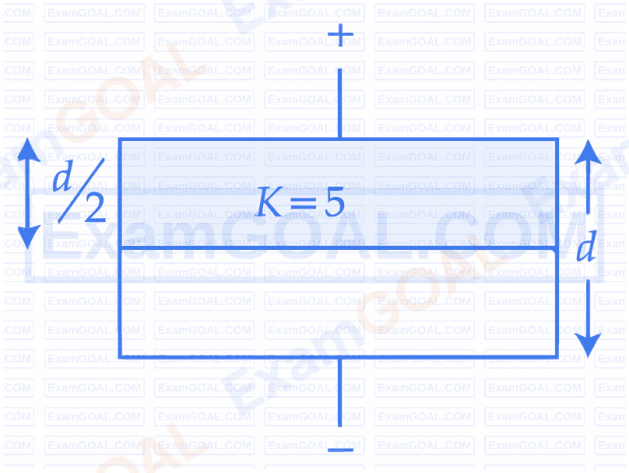
(A) -2 (B) 1 (C) -1 (D) 2

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Q55. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

A parallel plate air capacitor has a capacitance C. When it is half filled as shown in figure with a dielectric constant $K = 5$, the percentage increase in the capacitance is _____.



(A) 33.34 (B) 66.67 (C) 200 (D) 400

Q56. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Identify the correct statements :

A. Effective capacitance of a series combination of capacitors is always smaller than the smallest capacitance of the capacitor in the combination.

B. When a dielectric medium is placed between the charged plates of a capacitor, displacement of charges cannot occur due to insulation property of dielectric.

C. Increasing of area of capacitor plate or decreasing of thickness of dielectric is an alternate method to increase the capacitance.

D. For a point charge, concentric spherical shells centered at the location of the charge are equipotential surfaces.

Choose the **correct** answer from the options given below :

(A) A, C and D Only (B) A, B and C Only (C) B and D Only (D) C and D Only

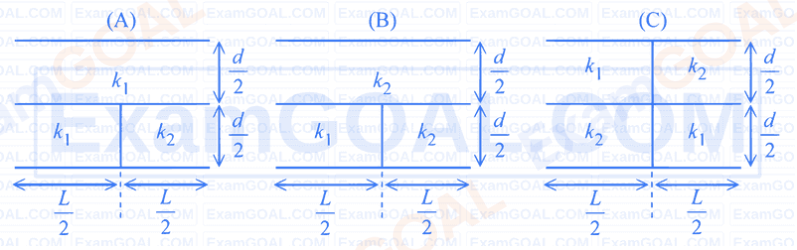
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Q57. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Three parallel plate capacitors each with area A and separation d are filled with two dielectrics (k_1 and k_2) in the following fashion. Which of the following is true?

$(k_1 > k_2)$



(A) $C_C > C_B > C_A$ (B) $C_B > C_C > C_A$ (C) $C_A > C_C > C_B$ (D) $C_C > C_A > C_B$

Q58. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

A parallel plate capacitor with plate separation 5 mm is charged by a battery. On introducing a mica sheet of 2 mm and maintaining the connections of the plates with the terminals of the battery, it is found that it draws 25% more charge from the battery. The dielectric constant of mica is ____

(A) 1.0 (B) 2.5 (C) 1.5 (D) 2.0

Q59. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

A parallel plate capacitor has capacitance C, when there is vacuum within the parallel plates. A sheet having thickness $\left(\frac{1}{3}\right)^{\text{rd}}$ of the separation between the plates and relative permittivity K is introduced between the plates. The new capacitance of the system is :

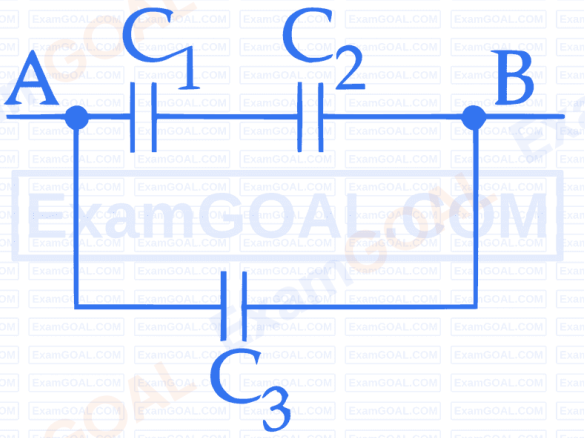
(A) $\frac{3CK^2}{(2K+1)^2}$ (B) $\frac{4KC}{3K-1}$ (C) $\frac{CK}{2+K}$ (D) $\frac{3KC}{2K+1}$

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Q60. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Three parallel plate capacitors C_1, C_2 and C_3 each of capacitance $5\mu\,\mathrm{F}$ are connected as shown in figure. The effective capacitance between points A and B, when the space between the parallel plates of C_1 capacitor is filled with a dielectric medium having dielectric constant of 4, is :



(A) $9\mu\,\mathrm{F}$ (B) $30\mu\,\mathrm{F}$ (C) $7.5\mu\,\mathrm{F}$ (D) $22.5\mu\,\mathrm{F}$

Q61. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Using a battery, a 100 pF capacitor is charged to 60 V and then the battery is removed. After that, a second uncharged capacitor is connected to the first capacitor in parallel. If the final voltage across the second capacitor is 20 V , its capacitance is: (in pF)

(A) 600 (B) 100 (C) 400 (D) 200

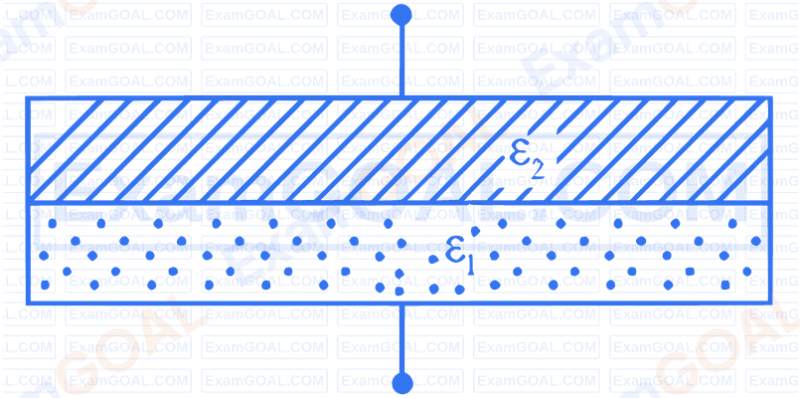
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Q62. JEE Main 2025 (Online) 3rd April Morning Shift

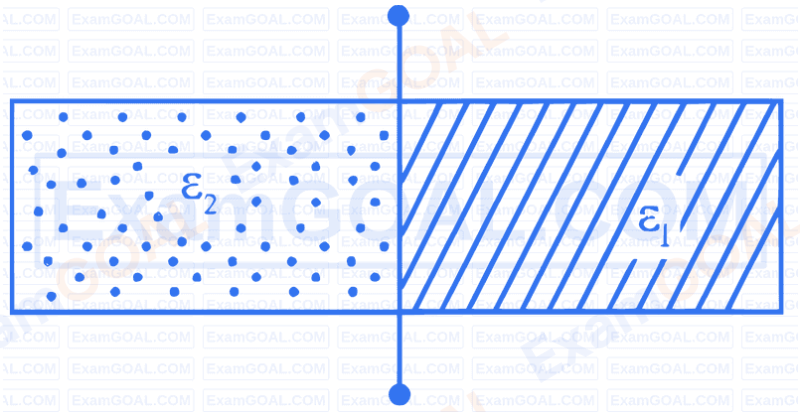
Open solution

A parallel plate capacitor is filled equally(half) with two dielectrics of dielectric constants ε_1 and ε_2, as shown in figures. The distance between the plates is d and area of each plate is A. If capacitance in first configuration and second configuration are C_1 and C_2 respectively, then $\frac{C_1}{C_2}$ is:

First Configuration



Second Configuration



(A) $\frac{\varepsilon_0(\varepsilon_1+\varepsilon_2)}{2}$

(B) $\frac{\varepsilon_1\varepsilon_2^2}{(\varepsilon_1+\varepsilon_2)^2}$

(C) $\frac{4\varepsilon_1\varepsilon_2}{(\varepsilon_1+\varepsilon_2)^2}$

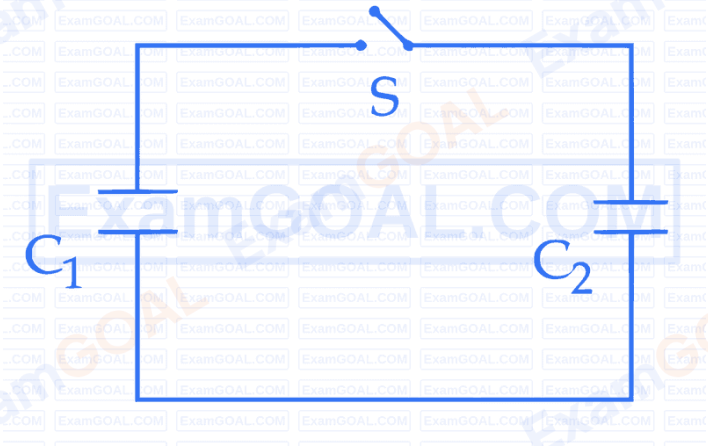
(D) $\frac{\varepsilon_1\varepsilon_2}{\varepsilon_1+\varepsilon_2}$

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Q63. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

A capacitor, $C_1 = 6\mu F$ is charged to a potential difference of $V_0 = 5V$ using a 5V battery. The battery is removed and another capacitor, $C_2 = 12\mu F$ is inserted in place of the battery. When the switch 'S' is closed, the charge flows between the capacitors for some time until equilibrium condition is reached. What are the charges (q_1 and q_2) on the capacitors C_1 and C_2 when equilibrium condition is reached.



(A) $q_1 = 10\mu C,\ q_2 = 20\mu C$

(B) $q_1 = 15\mu C,\ q_2 = 30\mu C$

(C) $q_1 = 20\mu C,\ q_2 = 10\mu C$

(D) $q_1 = 30\mu C,\ q_2 = 15\mu C$

Q64. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

A parallel plate capacitor of capacitance 1 μF is charged to a potential difference of 20 V. The distance between plates is 1 μm. The energy density between plates of capacitor is :

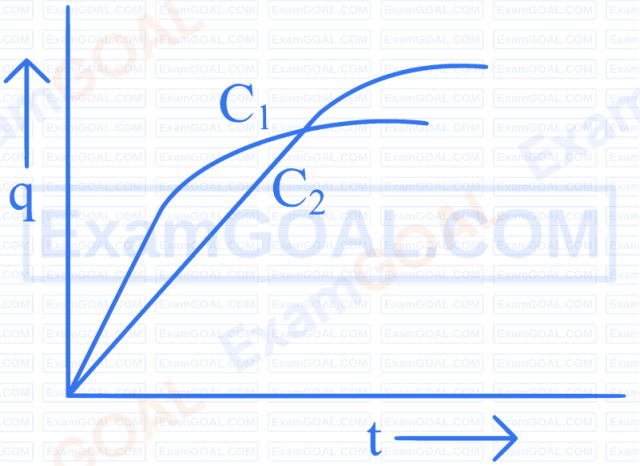
(A) 1.8×10^3 J/m^3 (B) 2×10^2 J/m^3 (C) 2×10^{-4} J/m^3 (D) 1.8×10^5 J/m^3

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Q65. JEE Main 2025 (Online) 28th January Morning Shift

Two capacitors C_1 and C_2 are connected in parallel to a battery. Charge-time graph is shown below for the two capacitors. The energy stored with them are U_1 and U_2, respectively. Which of the given statements is true?

Open solution



(A) $C_2 > C_1, U_2 > U_1$

(B) $C_1 > C_2, U_1 > U_2$

(C) $C_2 > C_1, U_2 < U_1$

(D) $C_1 > C_2, U_1 < U_2$

Q66. JEE Main 2025 (Online) 24th January Morning Shift

Consider a parallel plate capacitor of area A (of each plate) and separation ' d ' between the plates. If E is the electric field and ε_0 is the permittivity of free space between the plates, then potential energy stored in the capacitor is

Open solution

(A) $\varepsilon_0 E^2 Ad$

(B) $\frac{3}{4}\varepsilon_0 E^2 Ad$

(C) $\frac{1}{4}\varepsilon_0 E^2 Ad$

(D) $\frac{1}{2}\varepsilon_0 E^2 Ad$

Q67. JEE Main 2025 (Online) 24th January Morning Shift

A parallel plate capacitor was made with two rectangular plates, each with a length of $l = 3$ cm and breadth of $b = 1$ cm. The distance between the plates is 3μ m. Out of the following, which are the ways to increase the capacitance by a factor of 10 ?

Open solution

A. $l = 30$ cm, $b = 1$ cm, $d = 1\mu$ m

B. $l = 3$ cm, $b = 1$ cm, $d = 30\mu$ m

C. $l = 6$ cm, $b = 5$ cm, $d = 3\mu$ m

D. $l = 1$ cm, $b = 1$ cm, $d = 10\mu$ m

E. $l = 5$ cm, $b = 2$ cm, $d = 1\mu$ m

Choose the correct answer from the options given below:

(A) C only

(B) A only

(C) B and D only

(D) C and E only

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Q68. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Identify the valid statements relevant to the given circuit at the instant when the key is closed.



A. There will be no current through resistor R.

B. There will be maximum current in the connecting wires.

C. Potential difference between the capacitor plates A and B is minimum.

D. Charge on the capacitor plates is minimum.

Choose the correct answer from the options given below:

(A) A, C Only (B) C, D Only (C) A, B, D Only (D) B, C, D Only

Q69. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

An electron is made to enter symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The electron emerges out of the electric field region with a horizontal component of velocity $10^6\ \mathrm{m/s}$. If the magnitude of the electric field between the plates is $9.1\ \mathrm{V/cm}$, then the vertical component of velocity of electron is (mass of electron $= 9.1 \times 10^{-31}\ \mathrm{kg}$ and charge of electron $= 1.6 \times 10^{-19}\mathrm{C}$)

(A) $1 \times 10^6\ \mathrm{m/s}$ (B) $16 \times 10^6\ \mathrm{m/s}$

(C) $16 \times 10^4\ \mathrm{m/s}$ (D) 0

Q70. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

A parallel-plate capacitor of capacitance $40\mu\ \mathrm{F}$ is connected to a 100 V power supply. Now the intermediate space between the plates is filled with a dielectric material of dielectric constant $\mathrm{K} = 2$. Due to the introduction of dielectric material, the extra charge and the change in the electrostatic energy in the capacitor, respectively, are

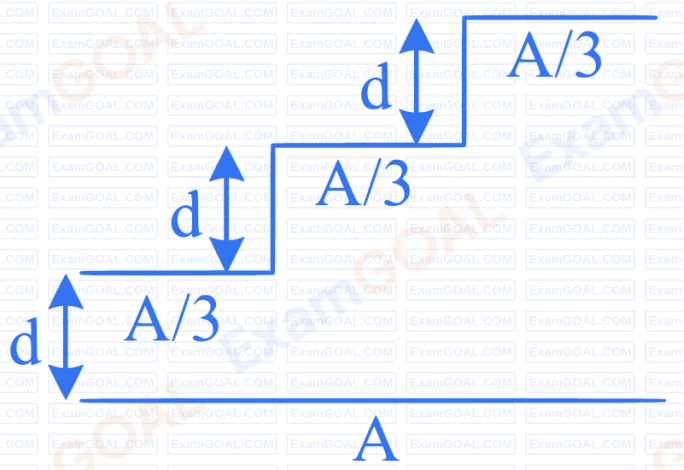
(A) 8 mC and 2.0 J (B) 4 mC and 0.2 J (C) 2 mC and 0.2 J (D) 2 mC and 0.4 J

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Q71. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

A capacitor is made of a flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is $\frac{A}{3}$ and the height is d, the capacitance of the arrangement is :



(A) $\frac{11\epsilon_o A}{20\,d}$

(B) $\frac{13\epsilon_o A}{17\,d}$

(C) $\frac{18\epsilon_o A}{11\,d}$

(D) $\frac{11\epsilon_o A}{18\,d}$

Q72. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

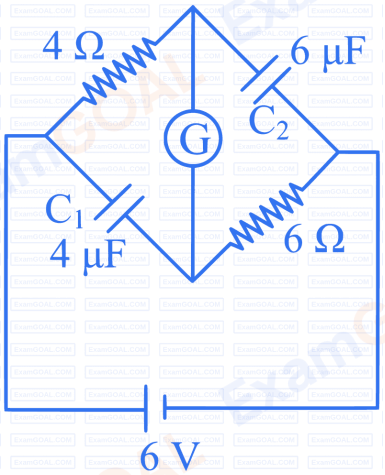
A capacitor has air as dielectric medium and two conducting plates of area $12\ \text{cm}^2$ and they are $0.6\ \text{cm}$ apart. When a slab of dielectric having area $12\ \text{cm}^2$ and $0.6\ \text{cm}$ thickness is inserted between the plates, one of the conducting plates has to be moved by $0.2\ \text{cm}$ to keep the capacitance same as in previous case. The dielectric constant of the slab is : (Given $\epsilon_0 = 8.834 \times 10^{-12}\ \text{F/m}$)

(A) 1.50 (B) 0.66 (C) 1.33 (D) 1

Q73. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

A galvanometer (G) of 2Ω resistance is connected in the given circuit. The ratio of charge stored in C_1 and C_2 is :



(A) 1 (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{1}{2}$

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Q74. JEE Main 2024 (Online) 1st February Morning Shift

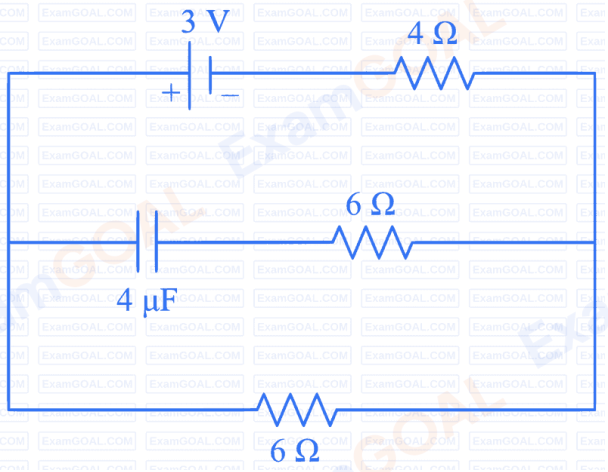
Two identical capacitors have same capacitance C. One of them is charged to the potential V and other to the potential $2\ \mathrm{V}$. The negative ends of both are connected together. When the positive ends are also joined together, the decrease in energy of the combined system is :

Open solution

(A) $\frac{1}{4}\mathrm{CV}^2$ (B) $\frac{3}{4}\mathrm{CV}^2$ (C) $\frac{1}{2}\mathrm{CV}^2$ (D) $2\mathrm{CV}^2$

Q75. JEE Main 2023 (Online) 13th April Evening Shift

In the network shown below, the charge accumulated in the capacitor in steady state will be:



Open solution

(A) 10.3 μC (B) 7.2 μC (C) 4.8 μC (D) 12 μC

Q76. JEE Main 2023 (Online) 11th April Evening Shift

A capacitor of capacitance C is charged to a potential V. The flux of the electric field through a closed surface enclosing the positive plate of the capacitor is :

Open solution

(A) Zero (B) $\frac{CV}{\varepsilon_0}$ (C) $\frac{CV}{2\varepsilon_0}$ (D) $\frac{2CV}{\varepsilon_0}$

Q77. JEE Main 2023 (Online) 11th April Morning Shift

A parallel plate capacitor of capacitance $2\ \mathrm{F}$ is charged to a potential V, The energy stored in the capacitor is E_1. The capacitor is now connected to another uncharged identical capacitor in parallel combination. The energy stored in the combination is E_2. The ratio $\mathrm{E}_2/\mathrm{E}_1$ is :

Open solution

(A) 1 : 2 (B) 2 : 3 (C) 2 : 1 (D) 1 : 4

Q78. JEE Main 2023 (Online) 10th April Evening Shift

The distance between two plates of a capacitor is d and its capacitance is C_1, when air is the medium between the plates. If a metal sheet of thickness $\frac{2d}{3}$ and of the same area as plate is introduced between the plates, the capacitance of the capacitor becomes C_2. The ratio $\frac{\mathrm{C}_2}{\mathrm{C}_1}$ is

Open solution

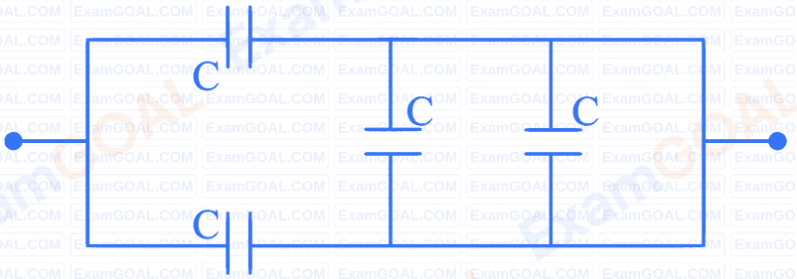
(A) 1 : 1 (B) 3 : 1 (C) 2 : 1 (D) 4 : 1

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Q79. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The equivalent capacitance of the combination shown is :

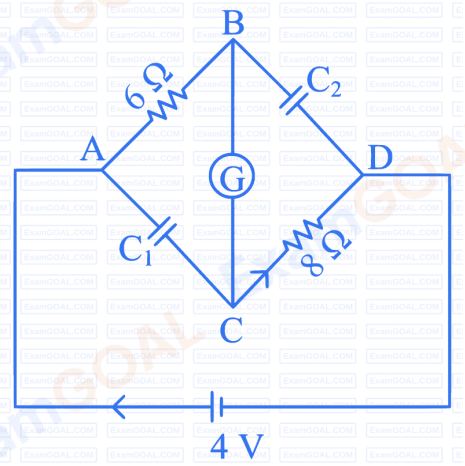


(A) $\frac{C}{2}$ (B) 4C (C) $\frac{5}{3}$C (D) 2C

Q80. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

In this figure the resistance of the coil of galvanometer G is $2\ \Omega$. The emf of the cell is $4\ \mathrm{V}$. The ratio of potential difference across C_1 and C_2 is:



(A) $\frac{5}{4}$ (B) $\frac{3}{4}$ (C) 1 (D) $\frac{4}{5}$

Q81. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

Given below are two statements: One is labeled as Assertion A and the other is labeled as Reason R.

Assertion A : Two metallic spheres are charged to the same potential. One of them is hollow and another is solid, and both have the same radii. Solid sphere will have lower charge than the hollow one.

Reason R : Capacitance of metallic spheres depends on the radii of spheres

In light of the above statements, choose the correct answer from the options given below.

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(D) $\mathbf{A}$ is true but $\mathbf{R}$ is false

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Q82. JEE Main 2023 (Online) 25th January Morning Shift

A parallel plate capacitor has plate area 40 cm^2 and plates separation 2 mm. The space between the plates is filled with a dielectric medium of a thickness 1 mm and dielectric constant 5. The capacitance of the system is :

Open solution

(A) $10\varepsilon_0\ \mathrm{F}$ (B) $24\varepsilon_0\ \mathrm{F}$

(C) $\frac{3}{10}\varepsilon_0\ \mathrm{F}$ (D) $\frac{10}{3}\varepsilon_0\ \mathrm{F}$

Q83. JEE Main 2022 (Online) 29th July Evening Shift

Two identical thin metal plates have charges q_1 and q_2 respectively such that $q_1 > q_2$. The plates were brought close to each other to form a parallel plate capacitor of capacitance C. The potential difference between them is :

Open solution

(A) $\frac{(q_1+q_2)}{C}$ (B) $\frac{(q_1-q_2)}{C}$

(C) $\frac{(q_1-q_2)}{2C}$ (D) $\frac{2(q_1-q_2)}{C}$

Q84. JEE Main 2022 (Online) 28th July Evening Shift

A slab of dielectric constant K has the same cross-sectional area as the plates of a parallel plate capacitor and thickness $\frac{3}{4}\ \mathrm{d}$, where d is the separation of the plates. The capacitance of the capacitor when the slab is inserted between the plates will be :

Open solution

(Given C_0 = capacitance of capacitor with air as medium between plates.)

(A) $\frac{4KC_0}{3+K}$ (B) $\frac{3KC_0}{3+K}$ (C) $\frac{3+K}{4KC_0}$ (D) $\frac{K}{4+K}$

Q85. JEE Main 2022 (Online) 28th July Morning Shift

Two capacitors, each having capacitance $40\ \mu\mathrm{F}$ are connected in series. The space between one of the capacitors is filled with dielectric material of dielectric constant K such that the equivalence capacitance of the system became $24\ \mu\mathrm{F}$. The value of K will be :

Open solution

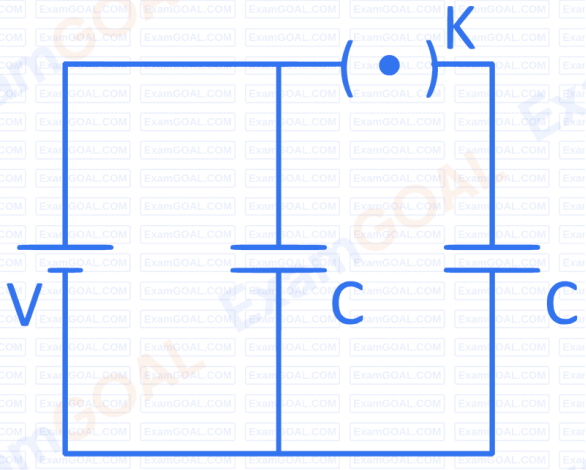
(A) 1.5 (B) 2.5 (C) 1.2 (D) 3

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Q86. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key 'K' is closed, the total energy stored across the combination is E_1. Now key 'K' is opened and dielectric of dielectric constant 5 is introduced between the plates of the capacitors. The total energy stored across the combination is now E_2. The ratio E_1/E_2 will be :



(A) $\frac{1}{10}$ (B) $\frac{2}{5}$ (C) $\frac{5}{13}$ (D) $\frac{5}{26}$

Q87. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

The total charge on the system of capacitors $C_1 = 1\mu\text{F}, C_2 = 2\mu\text{F}, \text{C}_3 = 4\mu\text{F}$ and $\text{C}_4 = 3\mu\text{F}$ connected in parallel is :

(Assume a battery of 20 V is connected to the combination)

(A) $200\,\mu\text{C}$ (B) 200 C (C) $10\,\mu\text{C}$ (D) 10 C

Q88. JEE Main 2022 (Online) 25th July Evening Shift

Open solution

Capacitance of an isolated conducting sphere of radius R_1 becomes n times when it is enclosed by a concentric conducting sphere of radius R_2 connected to earth. The ratio of their radii $\left(\frac{R_2}{R_1}\right)$ is :

(A) $\frac{n}{n-1}$ (B) $\frac{2n}{2n+1}$ (C) $\frac{n+1}{n}$ (D) $\frac{2n+1}{n}$

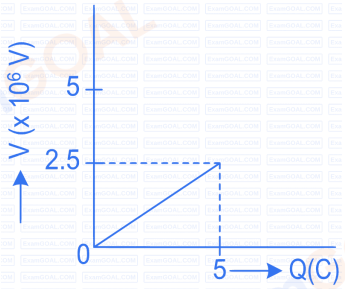
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Q89. JEE Main 2022 (Online) 25th July Morning Shift

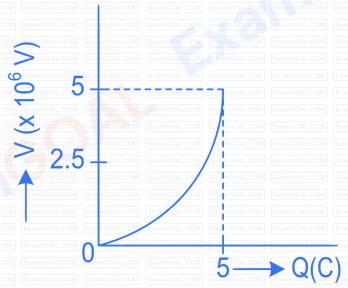
A condenser of $2\ \mu\mathrm{F}$ capacitance is charged steadily from 0 to $5\ \mathrm{C}$. Which of the following graph represents correctly the variation of potential difference (V) across it's plates with respect to the charge (Q) on the condenser?

Open solution

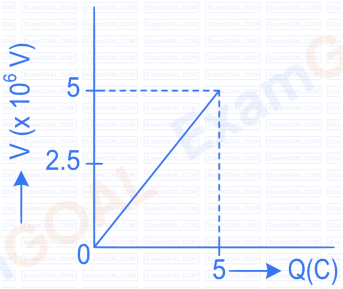
(A)



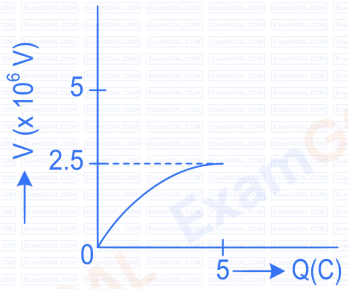
(B)



(C)



(D)



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Q90. JEE Main 2022 (Online) 30th June Morning Shift

Open solution

C_o is the capacitance of a parallel plate capacitor with air as a medium between the plates (as shown in Fig. 1). If half space between the plates is filled with a dielectric of relative permittivity ε_r (as shown in Fig. 2), the new capacitance of the capacitor will be :



Fig. 1



Fig. 2

(A) $\frac{C_o}{2}(1+\varepsilon_r)$

(B) $C_o+\varepsilon_r$

(C) $\frac{C_o\varepsilon_r}{2}$

(D) $C_o(1+\varepsilon_r)$

Q91. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

A capacitor is discharging through a resistor R. Consider in time t_1, the energy stored in the capacitor reduces to half of its initial value and in time t_2, the charge stored reduces to one eighth of its initial value. The ratio t_1/t_2 will be

(A) 1/2 (B) 1/3 (C) 1/4 (D) 1/6

Q92. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

A parallel plate capacitor filled with a medium of dielectric constant 10, is connected across a battery and is charged. The dielectric slab is replaced by another slab of dielectric constant 15. Then the energy of capacitor will :

(A) increase by 50% (B) decrease by 15% (C) increase by 25% (D) increase by 33%

Q93. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

A force of 10 N acts on a charged particle placed between two plates of a charged capacitor. If one plate of capacitor is removed, then the force acting on that particle will be.

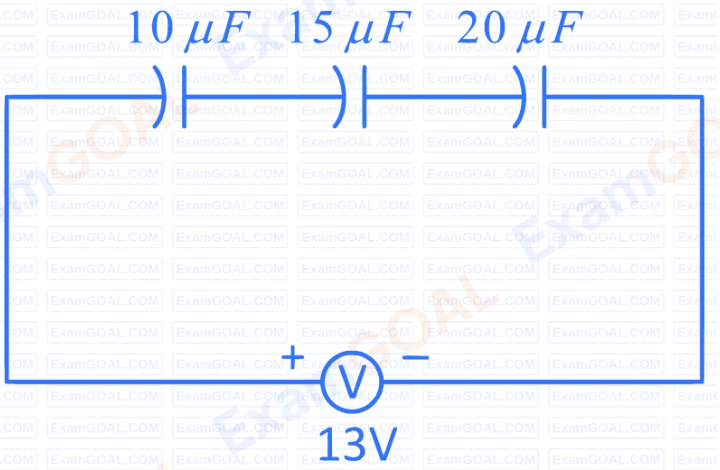
(A) 5 N (B) 10 N (C) 20 N (D) Zero

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Q94. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

The charge on capacitor of capacitance 15μF in the figure given below is :

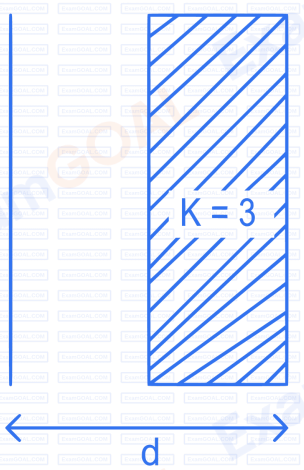


(A) 60μC (B) 130μC (C) 260μC (D) 585μC

Q95. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

A parallel plate capacitor with plate area A and plate separation d = 2 m has a capacitance of 4 μF. The new capacitance of the system if half of the space between them is filled with a dielectric material of dielectric constant K = 3 (as shown in figure) will be :



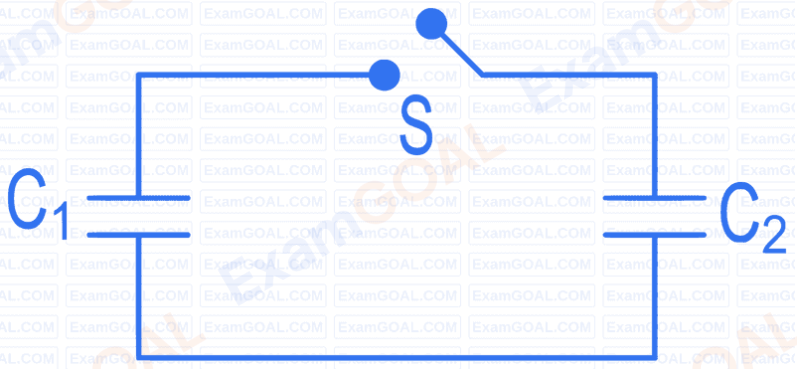
(A) 2μF (B) 32μF (C) 6μF (D) 8μF

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Q96. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

Two capacitors having capacitance C_1 and C_2 respectively are connected as shown in figure. Initially, capacitor C_1 is charged to a potential difference V volt by a battery. The battery is then removed and the charged capacitor C_1 is now connected to uncharged capacitor C_2 by closing the switch S. The amount of charge on the capacitor C_2, after equilibrium, is :



(A) $\frac{C_1C_2}{(C_1+C_2)}V$

(B) $\frac{(C_1+C_2)}{C_1C_2}V$

(C) $(C_1+C_2)V$

(D) $(C_1-C_2)V$

Q97. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

Two metallic plates form a parallel plate capacitor. The distance between the plates is 'd'. A metal sheet of thickness $\frac{d}{2}$ and of area equal to area of each plate is introduced between the plates. What will be the ratio of the new capacitance to the original capacitance of the capacitor?

(A) 2 : 1 (B) 1 : 2 (C) 1 : 4 (D) 4 : 1

Q98. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

If the charge on a capacitor is increased by 2 C, the energy stored in it increases by 44%. The original charge on the capacitor is (in C)

(A) 10 (B) 20 (C) 30 (D) 40

Q99. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

A parallel plate capacitor is formed by two plates each of area 30π cm^2 separated by 1 mm. A material of dielectric strength 3.6×10^7 Vm^{-1} is filled between the plates. If the maximum charge that can be stored on the capacitor without causing any dielectric breakdown is 7×10^{-6}C, the value of dielectric constant of the material is :

[Use $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ Nm^2 C^{-2}]

(A) 1.66 (B) 1.75 (C) 2.25 (D) 2.33

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Q100. JEE Main 2021 (Online) 1st September Evening Shift

A capacitor is connected to a 20 V battery through a resistance of 10Ω. It is found that the potential difference across the capacitor rises to 2 V in 1 μs. The capacitance of the capacitor is _____ μF. Given : $\ln\left(\frac{10}{9}\right) = 0.105$

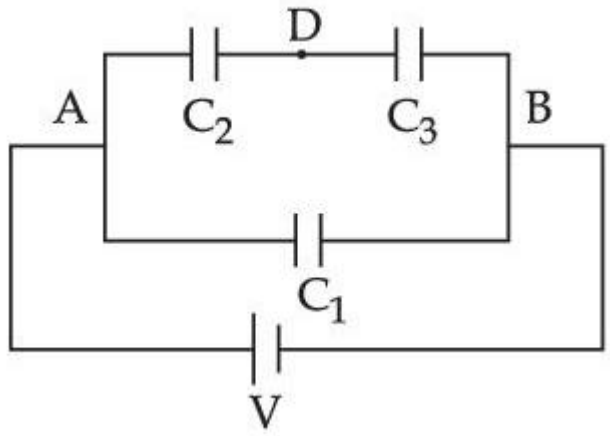
Open solution

(A) 9.52 (B) 0.95 (C) 0.105 (D) 1.85

Q101. JEE Main 2021 (Online) 27th August Evening Shift

Three capacitors C_1 = 2μF, C_2 = 6μF and C_3 = 12μF are connected as shown in figure. Find the ratio of the charges on capacitors C_1, C_2 and C_3 respectively :

Open solution

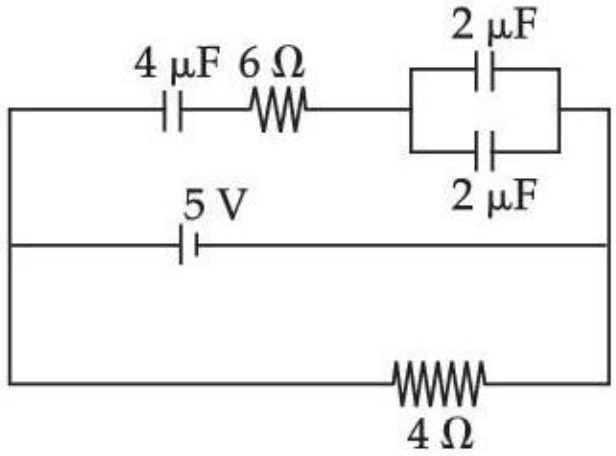


(A) 2 : 1 : 1 (B) 2 : 3 : 3 (C) 1 : 2 : 2 (D) 3 : 4 : 4

Q102. JEE Main 2021 (Online) 27th August Morning Shift

Calculate the amount of charge on capacitor of 4μF. The internal resistance of battery is 1Ω :

Open solution



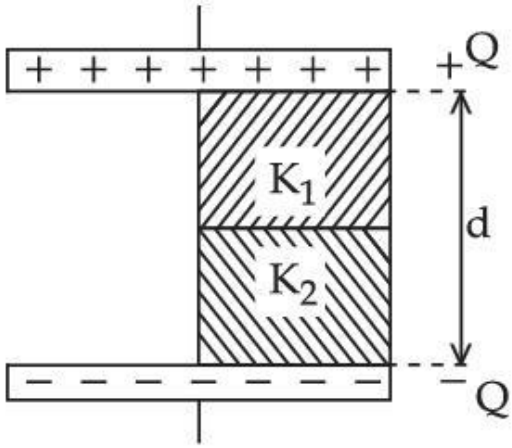
(A) 8 μC (B) zero (C) 16 μC (D) 4 μC

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Q103. JEE Main 2021 (Online) 26th August Evening Shift

Open solution

A parallel plate capacitor with plate area A has separation d between the plates. Two dielectric slabs of dielectric constant K_1 and K_2 of same area A/2 and thickness d/2 are inserted in the space between the plates. The capacitance of the capacitor will be given by:



(A) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1K_2}{K_1+K_2}\right)$

(B) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1K_2}{2(K_1+K_2)}\right)$

(C) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{K_1+K_2}{K_1K_2}\right)$

(D) $\frac{\varepsilon_0 A}{d}\left(\frac{1}{2}+\frac{2(K_1+K_2)}{K_1K_2}\right)$

Q104. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

The material filled between the plates of a parallel plate capacitor has resistivity 200 Ωm. The value of capacitance of the capacitor is 2 pF. If a potential difference of 40 V is applied across the plates of the capacitor, then the value of leakage current flowing out of the capacitor is : (given the value of relative permittivity of material is 50)

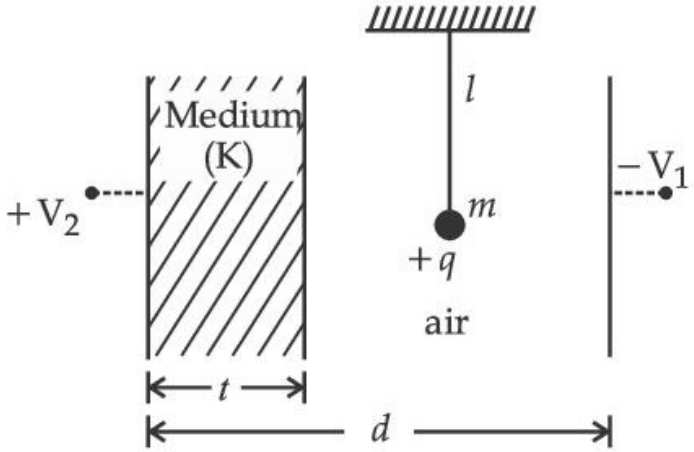
(A) 9.0 μA (B) 9.0 mA (C) 0.9 mA (D) 0.9 μA

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Q105. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

A simple pendulum of mass 'm', length 'l' and charge '+ q' suspended in the electric field produced by two conducting parallel plates as shown. The value of deflection of pendulum in equilibrium position will be



(A) $\tan^{-1}\left[\frac{q}{mg} \times \frac{C_1(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

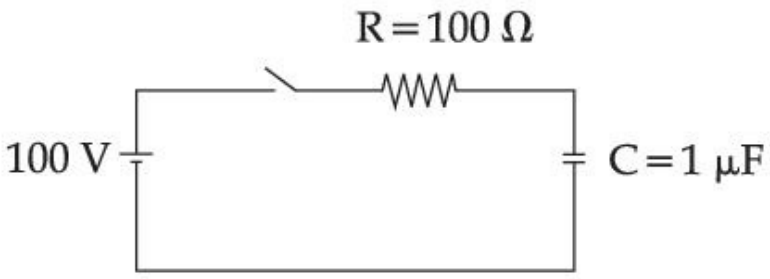
(B) $\tan^{-1}\left[\frac{q}{mg} \times \frac{C_2(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

(C) $\tan^{-1}\left[\frac{q}{mg} \times \frac{C_2(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

(D) $\tan^{-1}\left[\frac{q}{mg} \times \frac{C_1(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

Q106. JEE Main 2021 (Online) 27th July Morning Shift

Open solution



A capacitor of capacitance C = 1 μF is suddenly connected to a battery of 100 volt through a resistance R = 100 Ω. The time taken for the capacitor to be charged to get 50 V is :

[Take ln 2 = 0.69]

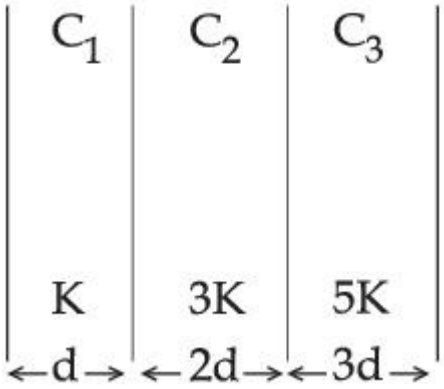
(A) 1.44×10^{-4} s (B) 3.33×10^{-4} s (C) 0.69×10^{-4} s (D) 0.30×10^{-4} s

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Q107. JEE Main 2021 (Online) 27th July Morning Shift

In the reported figure, a capacitor is formed by placing a compound dielectric between the plates of parallel plate capacitor. The expression for the capacity of the said capacitor will be :

(Given area of plate = A)



(A) $\frac{15}{34}\frac{K\varepsilon_0 A}{d}$

(B) $\frac{15}{6}\frac{K\varepsilon_0 A}{d}$

(C) $\frac{25}{6}\frac{K\varepsilon_0 A}{d}$

(D) $\frac{9}{6}\frac{K\varepsilon_0 A}{d}$

Open solution

Q108. JEE Main 2021 (Online) 27th July Morning Shift

Two capacitors of capacities 2C and C are joined in parallel and charged up to potential V. The battery is removed and the capacitor of capacity C is filled completely with a medium of dielectric constant K. The potential difference across the capacitors will now be :

(A) $\frac{V}{K+2}$

(B) $\frac{V}{K}$

(C) $\frac{3V}{K+2}$

(D) $\frac{3V}{K}$

Open solution

Q109. JEE Main 2021 (Online) 25th July Evening Shift

If q_f is the free charge on the capacitor plates and q_b is the bound charge on the dielectric slab of dielectric constant k placed between the capacitor plates, then bound charge q_b an be expressed as :

(A) $q_b = q_f\left(1-\frac{1}{\sqrt{k}}\right)$

(B) $q_b = q_f\left(1-\frac{1}{k}\right)$

(C) $q_b = q_f\left(1+\frac{1}{\sqrt{k}}\right)$

(D) $q_b = q_f\left(1+\frac{1}{k}\right)$

Open solution

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Q110. JEE Main 2021 (Online) 25th July Morning Shift

A parallel plate capacitor with plate area 'A' and distance of separation 'd' is filled with a dielectric. What is the capacity of the capacitor when permittivity of the dielectric varies as :

$\varepsilon(x) = \varepsilon_0 + kx$, for $\left(0 < x \leq \frac{d}{2}\right)$

$\varepsilon(x) = \varepsilon_0 + k(d - x)$, for $\left(\frac{d}{2} \leq x \leq d\right)$

(A) $\left(\varepsilon_0 + \frac{kd}{2}\right)^{2/kA}$

(B) $\frac{kA}{2\ln\left(\frac{2\varepsilon_0 + kd}{2\varepsilon_0}\right)}$

(C) 0

(D) $\frac{kA}{2}\ln\left(\frac{2\varepsilon_0}{2\varepsilon_0 - kd}\right)$

Open solution

Q111. JEE Main 2021 (Online) 16th March Morning Shift

For changing the capacitance of a given parallel plate capacitor, a dielectric material of dielectric constant K is used, which has the same area as the plates of the capacitor. The thickness of the dielectric slab is $\frac{3}{4}$d, where 'd' is the separation between the plates of parallel plate capacitor. The new capacitance (C') in terms of original capacitance (C_0) is given by the following relation :

(A) $C' = \frac{3+K}{4K}C_0$ (B) $C' = \frac{4+K}{3}C_0$ (C) $C' = \frac{4K}{K+3}C_0$ (D) $C' = \frac{4}{3+K}C_0$

Open solution

Q112. JEE Main 2021 (Online) 26th February Morning Shift

Consider the combination of 2 capacitors C_1 and C_2 with $C_2 > C_1$, when connected in parallel, the equivalent capacitance is $\frac{15}{4}$ times the equivalent capacitance of the same connected in series. Calculate the ratio of capacitors, $\frac{C_2}{C_1}$.

(A) $\frac{15}{11}$ (B) No Solutions (C) $\frac{29}{15}$ (D) $\frac{15}{4}$

Open solution

Q113. JEE Main 2021 (Online) 25th February Evening Shift

An electron with kinetic energy K_1 enters between parallel plates of a capacitor at an angle 'α' with the plates. It leaves the plates at angle 'β' with kinetic energy K_2. Then the ratio of kinetic energies $K_1 : K_2$ will be :

(A) $\frac{\cos^2\beta}{\cos^2\alpha}$ (B) $\frac{\cos\beta}{\cos\alpha}$

(C) $\frac{\sin^2\beta}{\cos^2\alpha}$ (D) $\frac{\cos\beta}{\sin\alpha}$

Open solution

Q114. JEE Main 2021 (Online) 24th February Morning Shift

Two equal capacitors are first connected in series and then in parallel. The ratio of the equivalent capacities in the two cases will be :

(A) 4 : 1 (B) 1 : 2 (C) 2 : 1 (D) 1 : 4

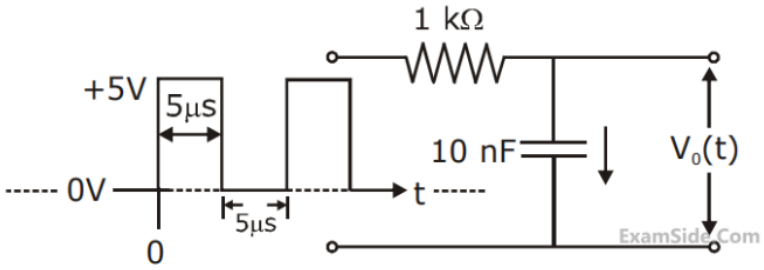
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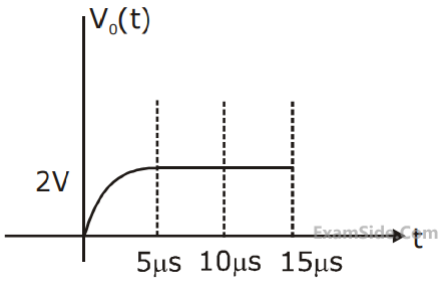
Q115. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

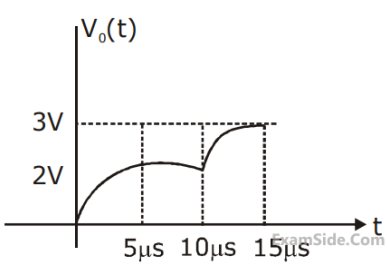
For the given input voltage waveform $V_{in}(t)$, the output voltage waveform $V_0(t)$, across the capacitor is correctly depicted by :



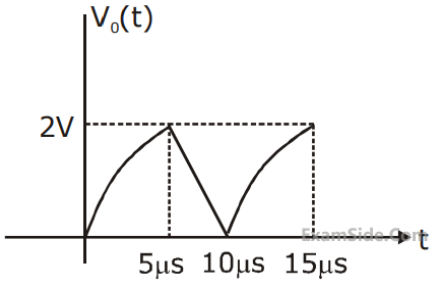
(A)



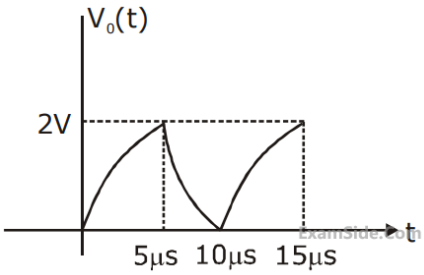
(B)



(C)



(D)



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Q116. JEE Main 2020 (Online) 5th September Evening Slot

A parallel plate capacitor has plate of length 'l', width 'w' and separation of plates is 'd'. It is connected to a battery of emf V. A dielectric slab of the same thickness 'd' and of dielectric constant k = 4 is being inserted between the plates of the capacitor. At what length of the slab inside plates, will the energy stored in the capacitor be two times the initial energy stored?

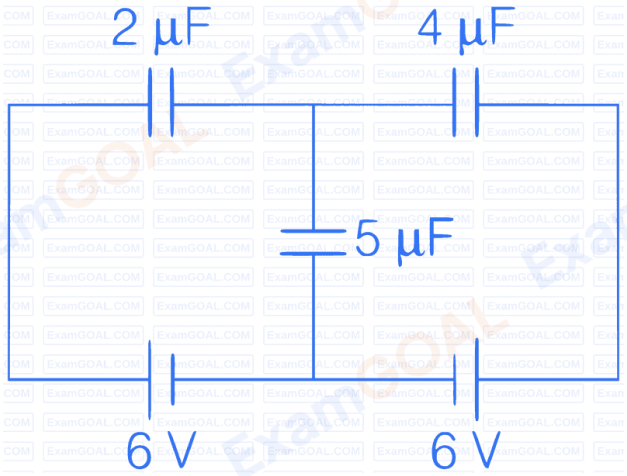
Open solution

(A) $\frac{l}{4}$ (B) $\frac{l}{2}$ (C) $\frac{2l}{3}$ (D) $\frac{l}{3}$

Q117. JEE Main 2020 (Online) 5th September Evening Slot

In the circuit shown, charge on the 5 μF capacitor is :

Open solution



(A) 5.45 μC (B) 18.00 μC (C) 10.90 μC (D) 16.36 μC

Q118. JEE Main 2020 (Online) 5th September Morning Slot

Two capacitors of capacitances C and 2C are charged to potential differences V and 2V, respectively. These are then connected in parallel in such a manner that the positive terminal of one is connected to the negative terminal of the other. The final energy of this configuration is :

Open solution

(A) Zero (B) $\frac{3}{2}CV^2$ (C) $\frac{9}{2}CV^2$ (D) $\frac{25}{6}CV^2$

Q119. JEE Main 2020 (Online) 4th September Evening Slot

A capacitor C is fully charged with voltage V_0. After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $\frac{C}{2}$. The energy loss in the process after the charge is distributed between the two capacitors is :

Open solution

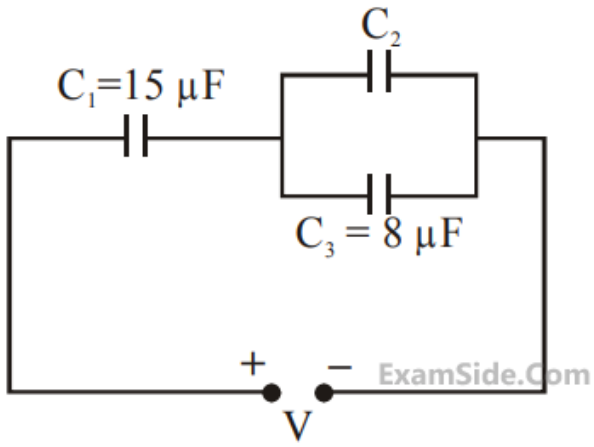
(A) $\frac{1}{2}CV_0^2$ (B) $\frac{1}{4}CV_0^2$ (C) $\frac{1}{3}CV_0^2$ (D) $\frac{1}{6}CV_0^2$

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Q120. JEE Main 2020 (Online) 3rd September Morning Slot

In the circuit shown in the figure, the total charge is 750 μC and the voltage across capacitor C_2 is 20 V. Then the charge on capacitor C_2 is :

Open solution



(A) 160 μC (B) 450 μC (C) 590 μC (D) 650 μC

Q121. JEE Main 2020 (Online) 2nd September Evening Slot

A 10 μF capacitor is fully charged to a potential difference of 50 V. After removing the source voltage it is connected to an uncharged capacitor in parallel. Now the potential difference across them becomes 20 V. The capacitance of the second capacitor is :

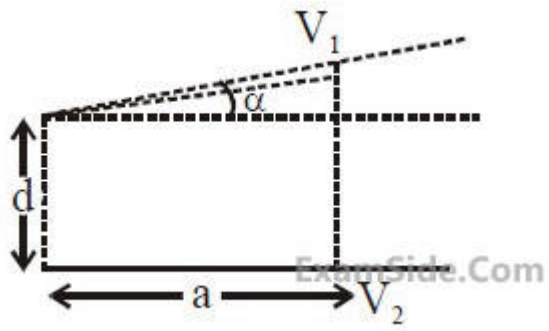
Open solution

(A) 20 μF (B) 15 μF (C) 10 μF (D) 30 μF

Q122. JEE Main 2020 (Online) 8th January Evening Slot

A capacitor is made of two square plates each of side 'a' making a very small angle α between them, as shown in figure. The capacitance will be close to :

Open solution



(A) $\frac{\varepsilon_0 a^2}{d}\left(1+\frac{\alpha a}{d}\right)$ (B) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{\alpha a}{4d}\right)$
(C) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{\alpha a}{2d}\right)$ (D) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{3\alpha a}{2d}\right)$

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Q123. JEE Main 2020 (Online) 8th January Morning Slot

Effective capacitance of parallel combination of two capacitors C_1 and C_2 is 10 μF. When these capacitors are individually connected to a voltage source of 1V, the energy stored in the capacitor C_2 is 4 times that of C_1. If these capacitors are connected in series, their effective capacitance will be :

Open solution

(A) 4.2 μF (B) 8.4 μF (C) 1.6 μF (D) 3.2 μF

Q124. JEE Main 2020 (Online) 7th January Morning Slot



Open solution

A parallel plate capacitor has plates of area A separated by distance 'd' between them. It is filled with a dielectric which has a dielectric constant that varies as k(x) = K(1 + αx) where 'x' is the distance measured from one of the plates. If (ad) << 1, the total capacitance of the system is best given by the expression :

(A) $\frac{A\in_0 K}{d}\left(1+\left(\frac{\alpha d}{2}\right)^2\right)$

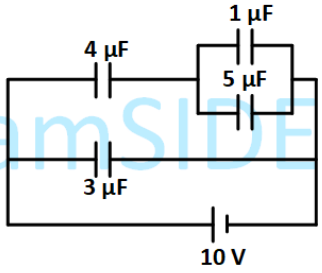
(B) $\frac{A\in_0 K}{d}\left(1+\frac{\alpha d}{2}\right)$

(C) $\frac{A\in_0 K}{d}\left(1+\frac{\alpha^2 d^2}{2}\right)$

(D) $\frac{A\in_0 K}{d}(1+\alpha d)$

Q125. JEE Main 2019 (Online) 12th April Evening Slot

In the given circuit, the charge on 4 μF capacitor will be :



Open solution

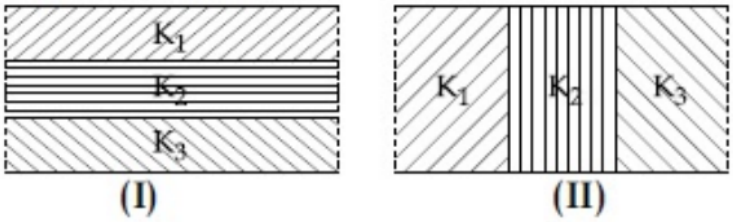
(A) 5.4 μC (B) 9.6 μC (C) 13.4 μC (D) 24 μC

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Q126. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

Two identical parallel plate capacitors, of capacitance C each, have plates of area A, separated by a distance d. The space between the plates of the two capacitors, is filled with three dielectrics, of equal thickness and dielectric constants K_1, K_2 and K_3. The first capacitor is filled as shown in fig.I, and the second one is filled as shown in fig II. If these two modified capacitors are charged by the same potential V, the ratio of the energy stored in the two, would be (E_1 refers to capacitor (I) and E_2 to capacitor (II)):



(A) $\frac{E_1}{E_2} = \frac{(K_1+K_2+K_3)(K_2K_3+K_3K_1+K_1K_2)}{K_1K_2K_3}$

(B) $\frac{E_1}{E_2} = \frac{K_1K_2K_3}{(K_1+K_2+K_3)(K_2K_3+K_3K_1+K_1K_2)}$

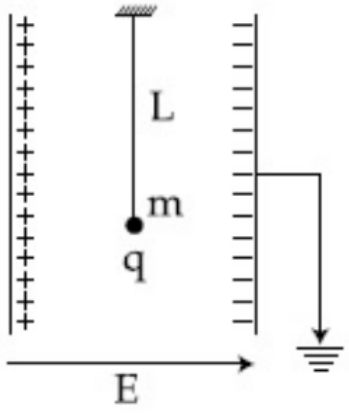
(C) $\frac{E_1}{E_2} = \frac{(K_1+K_2+K_3)(K_2K_3+K_3K_1+K_1K_2)}{9K_1K_2K_3}$

(D) $\frac{E_1}{E_2} = \frac{9K_1K_2K_3}{(K_1+K_2+K_3)(K_2K_3+K_3K_1+K_1K_2)}$

Q127. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E, as shown in figure. Its bob has mass m and charge q. The time period of the pendulum is given by :



(A) $2\pi\sqrt{\frac{L}{\sqrt{g^2-\frac{q^2E^2}{m^2}}}}$

(B) $2\pi\sqrt{\frac{L}{\left(g+\frac{qE}{m}\right)}}$

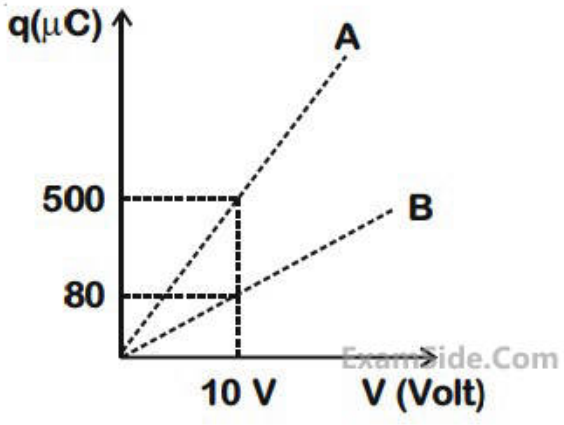
(C) $2\pi\sqrt{\frac{L}{\sqrt{g^2+\frac{q^2E^2}{m^2}}}}$

(D) $2\pi\sqrt{\frac{L}{\left(g-\frac{qE}{m}\right)}}$

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Q128. JEE Main 2019 (Online) 10th April Morning Slot

Figure shows charge (q) versus voltage (V) graph for series and parallel combination of two given capacitors. The capacitances are :



Open solution

(A) 40 μF and 10 μF
(B) 60 μF and 40 μF
(C) 20 μF and 30 μF
(D) 50 μF and 30 μF

Q129. JEE Main 2019 (Online) 9th April Evening Slot

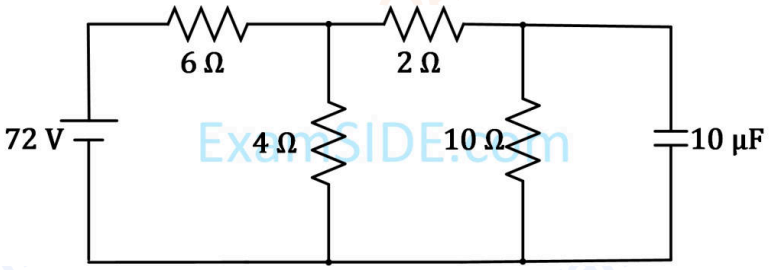
The parallel combination of two air filled parallel plate capacitors of capacitance C and nC is connected to a battery of voltage, V. When the capacitors are fully charged, the battery is removed and after that a dielectric material of dielectric constant K is placed between the two plates of the first capacitor. The new potential difference of the combined system is :-

Open solution

(A) V
(B) $\frac{V}{K+n}$
(C) $\frac{(n+1)V}{K+n}$
(D) $\frac{nV}{K+n}$

Q130. JEE Main 2019 (Online) 9th April Morning Slot

Determine the charge on the capacitor in the following circuit :



Open solution

(A) 200μC
(B) 2μC
(C) 10μC
(D) 60μC

Q131. JEE Main 2019 (Online) 9th April Morning Slot

A capacitor with capacitance 5μF is charged to 5μC. If the plates are pulled apart to reduce the capacitance to 2μF, how much work is done ?

Open solution

(A) 2.16×10^{-6} J
(B) 2.55×10^{-6} J
(C) 3.75×10^{-6} J
(D) 6.25×10^{-6} J

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Q132. JEE Main 2019 (Online) 8th April Evening Slot

A parallel plate capacitor has 1μF capacitance. One of its two plates is given +2μC charge and the other plate, +4μC charge. The potential difference developed across the capacitor is:-

(A) 1V (B) 5V (C) 2V (D) 3V

Open solution

Q133. JEE Main 2019 (Online) 8th April Morning Slot

Voltage rating of a parallel plate capacitor is 500V. Its dielectric can withstand a maximum electric field of 10^6 V/m. The plate area is 10^{-4} m^2. What is the dielectric constant if the capacitance is 15 pF? (given $\varepsilon_0 = 8.86 \times 10^{-12}$ C^2/Nm^2)

(A) 8.5 (B) 4.5 (C) 3.8 (D) 6.2

Open solution

Q134. JEE Main 2019 (Online) 12th January Evening Slot

A parallel plate capacitor with plates of area 1 m^2 each, are at a separation of 0.1 m. If the electric field between the plates is 100 N/C, the magnitude of charge on each plate is :

(Take $\varepsilon_0 = 8.85 \times 10^{-12}$ $\frac{C^2}{N-m^2}$)

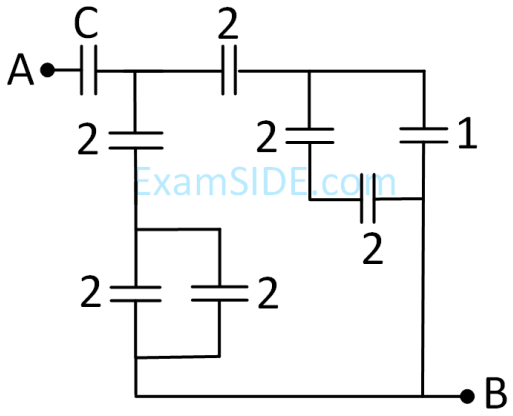
(A) 9.85×10^{-10} C (B) 8.85×10^{-10} C

(C) 6.85×10^{-10} C (D) 7.85×10^{-10} C

Open solution

Q135. JEE Main 2019 (Online) 12th January Evening Slot

In the circuit shown, find C if the effective capacitance of the whole circuit is to be 0.5 μF. All values in the circuit are in μF.



(A) $\frac{6}{5}\mu F$ (B) $\frac{7}{11}\mu F$ (C) 4μF (D) $\frac{7}{10}\mu F$

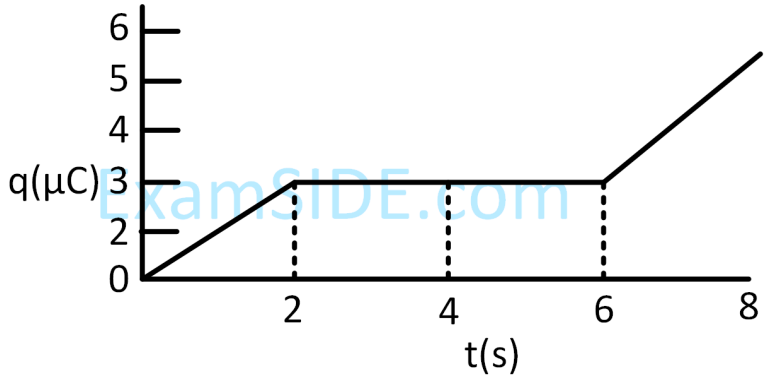
Open solution

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Q136. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

The charge on a capacitor plate in a circuit, as a function of time, is shown in the figure :



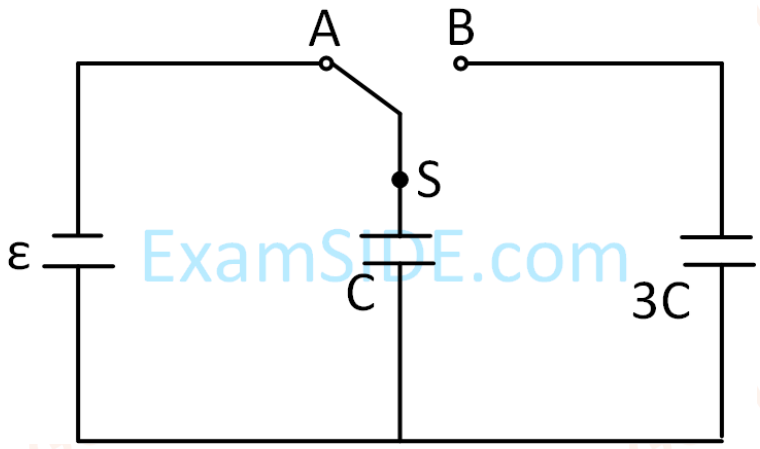
What is the value of current at t = 4 s ?

(A) zero (B) 1.5 μA (C) 2 μA (D) 3 μA

Q137. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

In the figure shown, after the switch 'S' is turned from position 'A' to position 'B', the energy dissipated in the circuit in terms of capacitance 'C' and total charge 'Q' is :



(A) $\frac{1}{8}\frac{Q^2}{C}$ (B) $\frac{5}{8}\frac{Q^2}{C}$ (C) $\frac{3}{4}\frac{Q^2}{C}$ (D) $\frac{3}{8}\frac{Q^2}{C}$

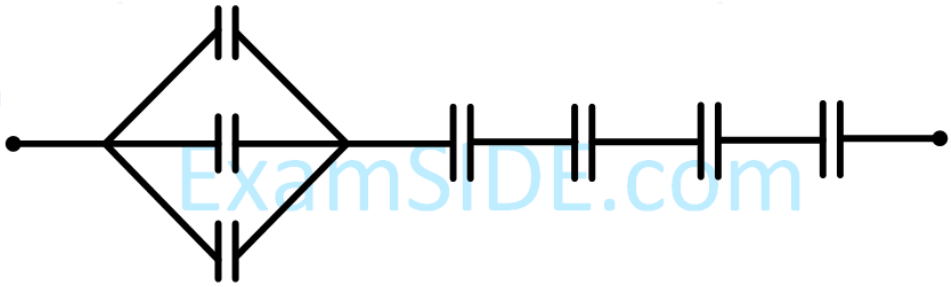
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Q138. JEE Main 2019 (Online) 11th January Evening Slot

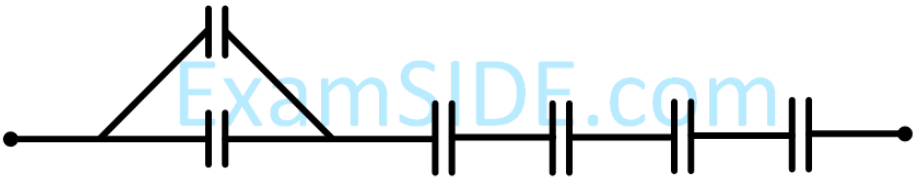
Seven capacitors, each of capacitance 2 μF, are to be connected in a configuration to obtain an effective capacitance of $\left(\frac{6}{13}\right)\mu F$. Which of the combinations, shown in figures below, will achieve the desired value

Open solution

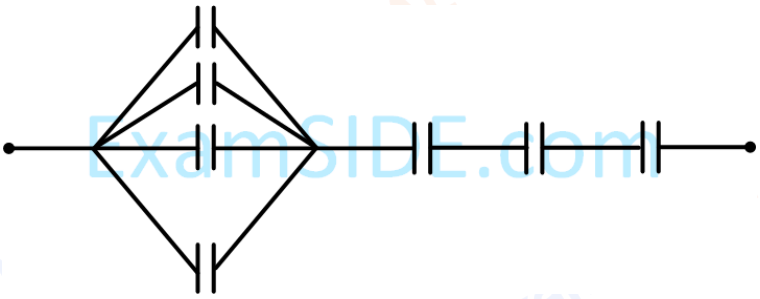
(A)



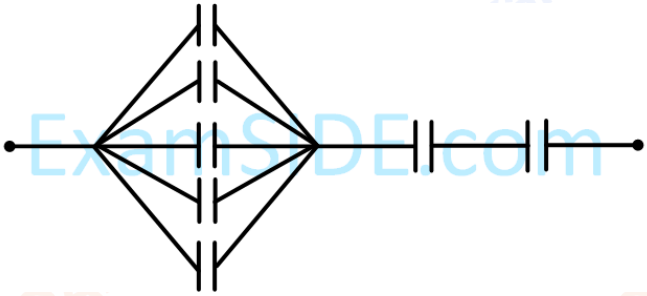
(B)



(C)



(D)

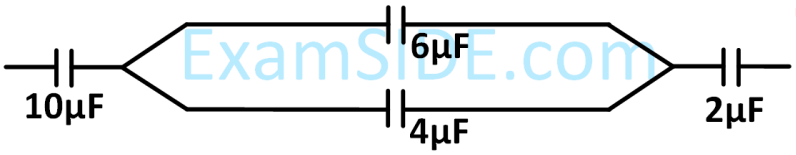


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Q139. JEE Main 2019 (Online) 11th January Morning Slot

In the figure shown below, the charge on the left plate of the 10μF capacitor is –30μC. The charge on the right plate of the 6 μF capacitor is :

Open solution



(A) + 12 μC (B) + 18 μC (C) − 18 μC (D) − 12 μC

Q140. JEE Main 2019 (Online) 10th January Evening Slot

A parallel plate capacitor having capacitance 12 pF is charged by a battery to a potential difference of 10 V between its plates. The charging battery is now disconnected and a porcelain slab of dielectric constant 6.5 is slipped between the plates. The work done by the capacitor on the slab is :

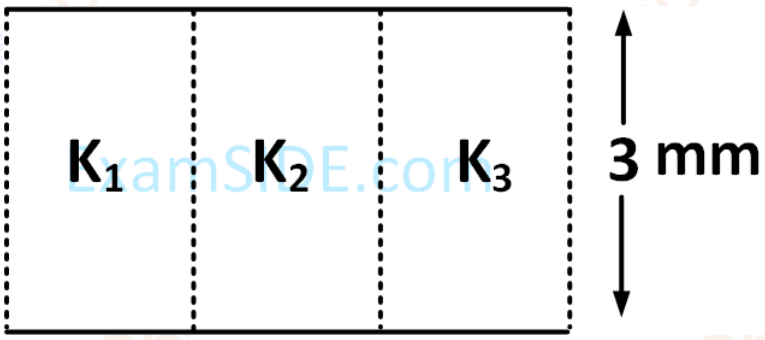
Open solution

(A) 508 pJ (B) 692 pJ (C) 560 pJ (D) 600 pJ

Q141. JEE Main 2019 (Online) 10th January Morning Slot

A parallel plate capacitor is of area 6 cm^2 and a separation 3 mm. The gap is filled with three dielectric materials of equal thickness (see figure) with dielectric constants K_1 = 10, K_2 = 12 and K_3 = 14. The dielectric constant of a material which when fully inserted in above capacitor, gives same capacitance would be -

Open solution



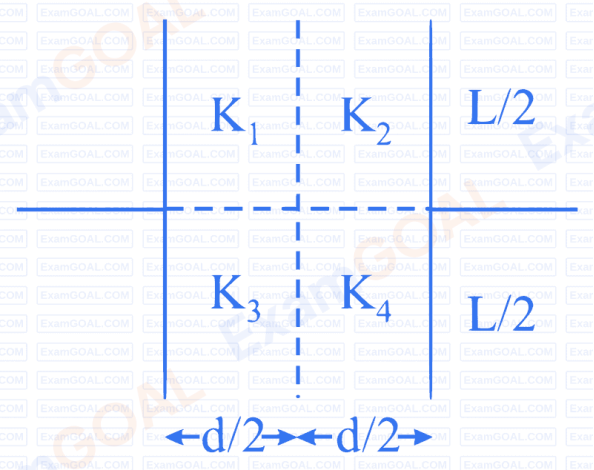
(A) 12 (B) 36 (C) 14 (D) 4

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Q142. JEE Main 2019 (Online) 9th January Evening Slot

A parallel plate capacitor with square plates is filled with four dielectrics of dielectric constants K_1, K_2, K_3, K_4 arranged as shown in the figure. The effective dielectric constant K will be :

Open solution



(A) $K = \frac{K_1K_2}{K_1+K_2} + \frac{K_3K_4}{K_3+K_4}$

(B) $K = \frac{(K_1+K_2)(K_3+K_4)}{2(K_1+K_2+K_3+K_4)}$

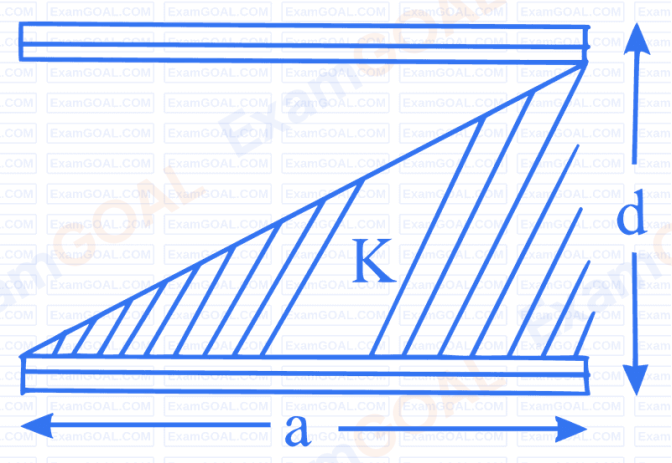
(C) $K = \frac{(K_1+K_2)(K_3+K_4)}{K_1+K_2+K_3+K_4}$

(D) $K = \frac{(K_1+K_4)(K_2+K_3)}{2(K_1+K_2+K_3+K_4)}$

Q143. JEE Main 2019 (Online) 9th January Morning Slot

A parallel plate capacitor is made of two square plates of side 'a', separated by a distance d (d < < a). The lower triangular portion is filled with a dielectric of dielectric constant K, as shown in the figure. Capacitance of this capacitor is :

Open solution



(A) $\frac{K\epsilon_0a^2}{2d(K+1)}$

(B) $\frac{K\epsilon_0a^2}{d(K-1)} \ln K$

(C) $\frac{K\epsilon_0a^2}{d} \ln K$

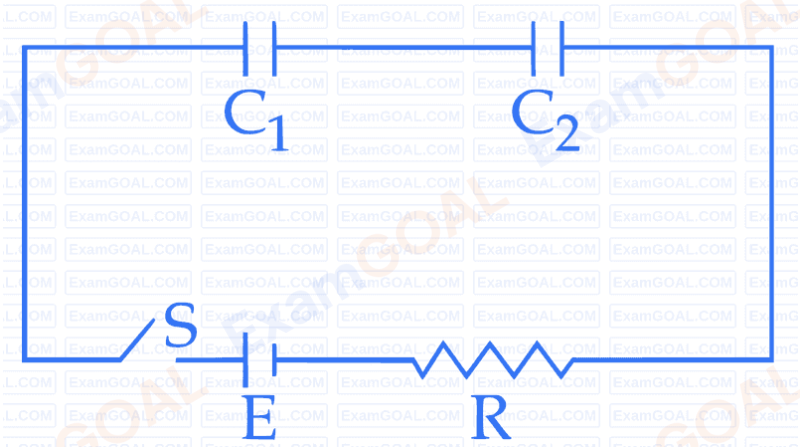
(D) $\frac{1}{2}\frac{K\epsilon_0a^2}{d}$

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Q144. JEE Main 2018 (Online) 16th April Morning Slot

In the following circuit, the switch S is closed at t = 0. The charge on the capacitor C_1 as a function of time will be given by $\left(C_{eq} = \frac{C_1C_2}{C_1+C_2}\right)$

Open solution



(A) $C_1E\left[1-\exp\left(-tR/C_1\right)\right]$
(B) $C_2E\left[1-\exp\left(-t/RC_2\right)\right]$
(C) $C_{eq}E\left[1-\exp\left(-t/RC_{eq}\right)\right]$
(D) $C_{eq}E\ \exp\left(-t/RC_{eq}\right)$

Q145. JEE Main 2018 (Offline)

A parallel plate capacitor of capacitance 90 pF is connected to a battery of emf 20 V. If a dielectric material of dielectric constant K = 5/3 is inserted between the plates, the magnitude of the induced charge will be :

Open solution

(A) 0.9 n C
(B) 1.2 n C
(C) 0.3 n C
(D) 2.4 n C

Q146. JEE Main 2018 (Online) 15th April Evening Slot

A parallel plate capacitor with area 200 cm^2 and separation between the plates 1.5 cm, is connected across a battery of emf V. If the force of attraction between the plates is $25\times10^{-6}N$, the value of V is approximately : $\left(\epsilon_o = 8.85\times10^{-12}\frac{C^2}{N.m^2}\right)$

Open solution

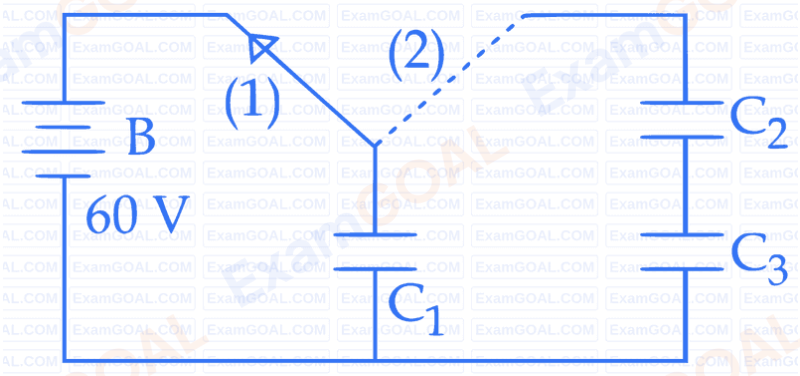
(A) 250 V
(B) 100 V
(C) 300 V
(D) 150 V

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Q147. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

A capacitor C_1 = 1.0 μF is charged up to a voltage V = 60 V by connecting it to battery B through switch (1). Now C_1 is disconnected from battery and connected to a circuit consisting of two uncharged capacitors $C_2 = 3.0\mu F$ and C_3 = 6.0 μF through switch (2), as shown in the figure. The sum of final charges on C_2 and C_3 is :

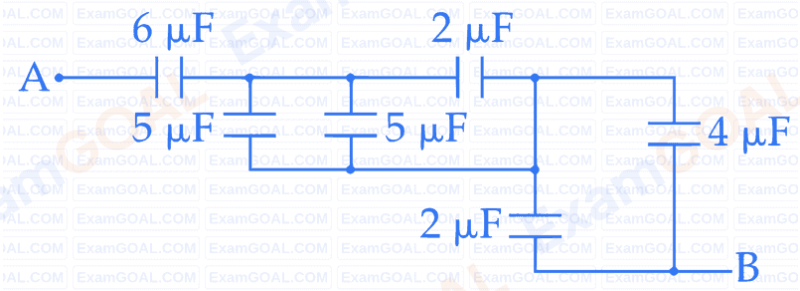


(A) 40 μC (B) 36 μC (C) 20 μC (D) 54 μC

Q148. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

The equivalent capacitance between A and B in the circuit given below, is :



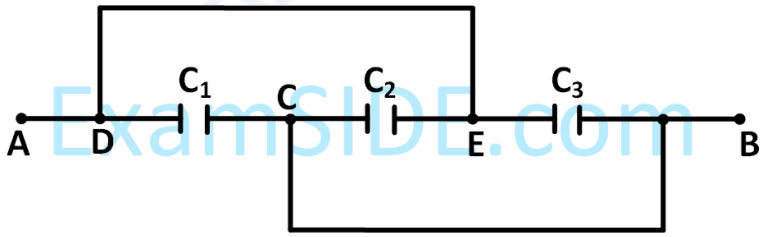
(A) $2.4\,\mu F$ (B) $4.9\,\mu F$ (C) $3.6\,\mu F$ (D) $5.4\,\mu F$

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Q149. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

A combination of parallel plate capacitors is maintained at a certain potential difference.



When a 3 mm thick slab is introduced between all the plates, in order to maintain the same potential difference, the distance between the plates is increased by 2.4 mm. Find the dielectric constant of the slab.

(A) 3 (B) 4 (C) 5 (D) 6

Q150. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

The energy stored in the electric field produced by a metal sphere is 4.5 J. If the sphere contains 4 μC charge, its radius will be :

[Take : $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ N $-$ m^2/C^2]

(A) 20 mm (B) 32 mm (C) 28 mm (D) 16 mm

Q151. JEE Main 2017 (Offline)

Open solution

A capacitance of 2 μF is required in an electrical circuit across a potential difference of 1.0 kV. A large number of 1 μF capacitors are available which can withstand a potential difference of not more than 300 V. The minimum number of capacitors required to achieve this is:

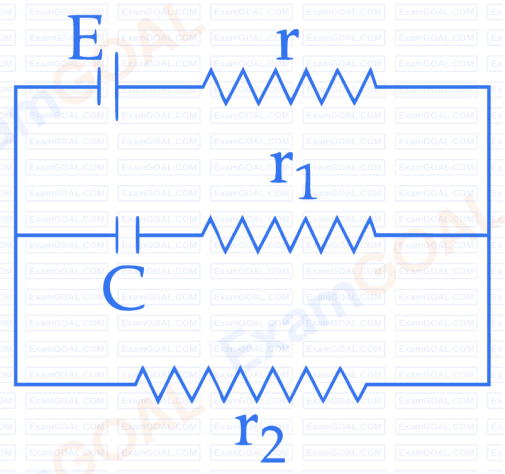
(A) 2 (B) 16 (C) 32 (D) 24

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Q152. JEE Main 2017 (Offline)

In the given circuit diagram when the current reaches steady state in the circuit, the charge on the capacitor of capacitance C will be:

Open solution

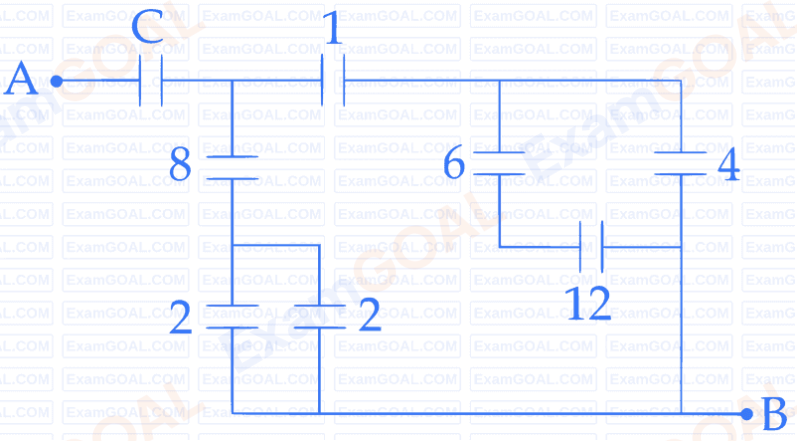


(A) $CE\frac{r_1}{(r_1+r)}$ (B) CE (C) $CE\frac{r_1}{(r_2+r)}$ (D) $CE\frac{r_2}{(r+r_2)}$

Q153. JEE Main 2016 (Online) 10th April Morning Slot

Figure shows a network of capacitors where the numbers indicates capacitances in micro Farad. The value of capacitance C if the equivalent capacitance between point A and B is to be 1 μF is :

Open solution



(A) $\frac{31}{23}\ \mu F$ (B) $\frac{32}{23}\ \mu F$ (C) $\frac{33}{23}\ \mu F$ (D) $\frac{34}{23}\ \mu F$

Q154. JEE Main 2016 (Online) 9th April Morning Slot

Three capacitors each of 4 μF are to be connected in such a way that the effective capacitance is 6 μF. This can be done by connecting them :

Open solution

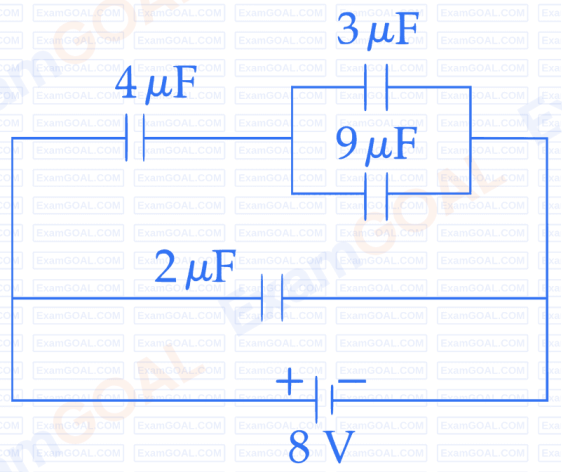
(A) all in series
(B) two in series and one in parallel
(C) all in parallel
(D) two in parallel and one in series

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Q155. JEE Main 2016 (Offline)

A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the $4\ \mu F$ and $9\ \mu F$ capacitors), at a point distance $30\ m$ from it, would equal :

Open solution

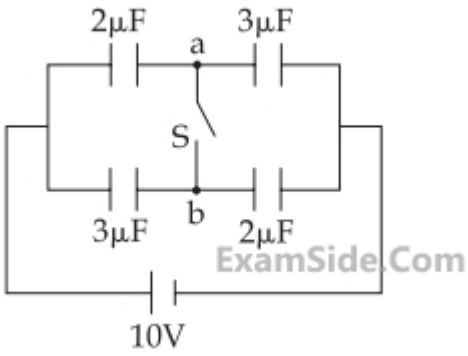


(A) $420N/C$ (B) $480N/C$ (C) $240N/C$ (D) $360N/C$

Q156. JEE Main 2015 (Online) 11th April Morning Slot

In figure is shown a system of four capacitors connected across a 10 V battery. Charge that flow from switch S when it is closed is :

Open solution



(A) 5 μC from b to a (B) 20 μC from a to b

(C) 5 μC from a to b (D) zero

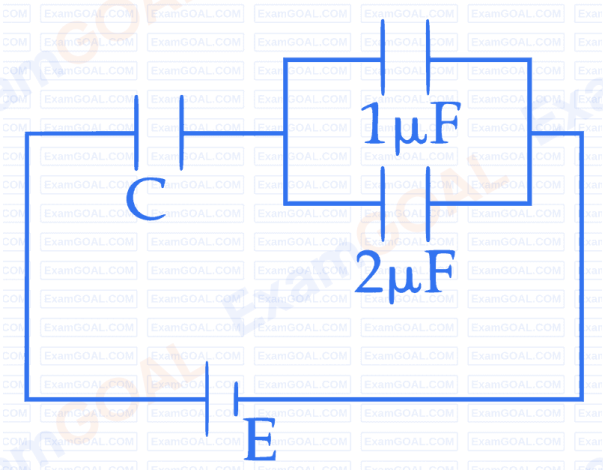
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Q157. JEE Main 2015 (Offline)

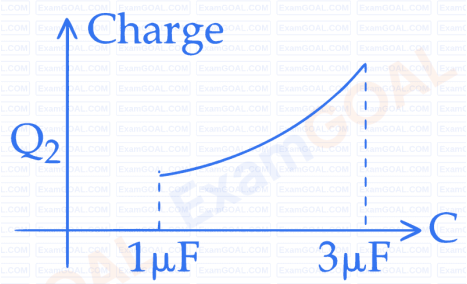
In the given circuit, charges Q_2 on the $2\mu F$ capacitor changes as C is varied from $1\ \mu F$ to $3\mu F$. Q_2 as a function of 'C' is given properly by:

($figures\ are\ drawn\ schematically\ and\ are\ not\ to\ scale$)

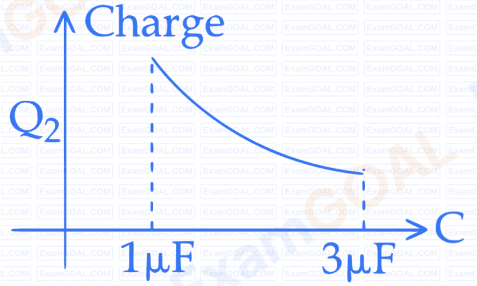
Open solution



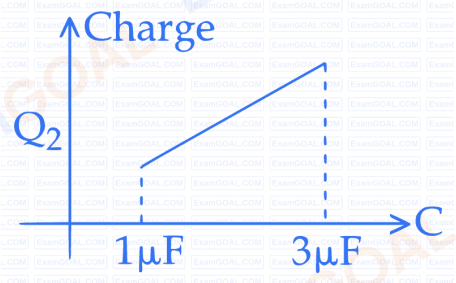
(A)



(B)

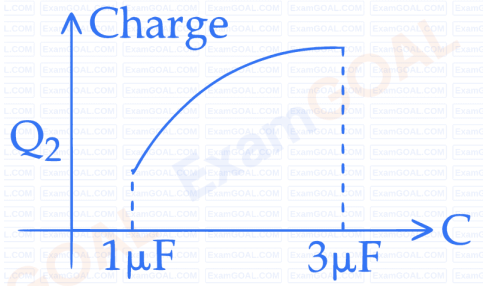


(C)



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(D)



Q158. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

The gap between the plates of a parallel plate capacitor of area A and distance between plates d, is filled with a dielectric whose permittivity varies linearly from $\in_1$ at one plate to $\in_2$ at the other. The capacitance of capacitor is :

(A) $\in_0(\in_1 + \in_2)A/d$

(B) $\in_0(\in_2 + \in_1)A/2d$

(C) $\in_0 A/[d \ln(\in_2/\in_1)]$

(D) $\in_0(\in_2 - \in_1)A/[d \ln(\in_2/\in_1)]$

Q159. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

The space between the plates of a parallel plate capacitor is filled with a 'dielectric' whose 'dielectric constant' varies with distance as per the relation :

$K(x) = K_0 + \lambda x$ (λ = a constant)

The capacitance C, of this capacitor, would be related to its 'vacuum' capacitance C_0 as per relation :

(A) $C = \frac{\lambda d}{\ln(1+K_0\lambda d)} C_0$

(B) $C = \frac{\lambda}{d.\ln(1+K_0\lambda d)} C_0$

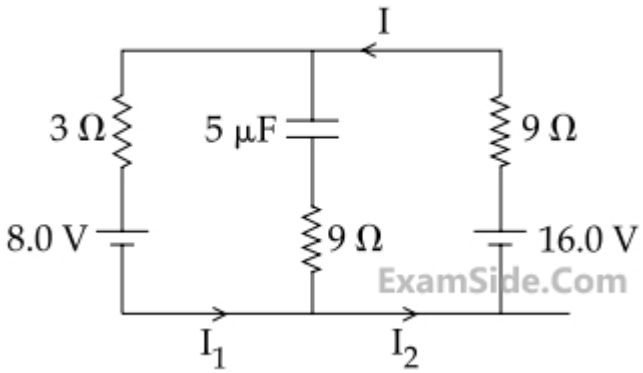
(C) $C = \frac{\lambda d}{\ln(1+\lambda d/K_0)} C_0$

(D) $C = \frac{\lambda}{d.\ln(1+K_0/\lambda d)} C_0$

Q160. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

The circuit shown here has two batteries of 8.0 V and 16.0 V and three resistors 3 Ω, 9 Ω and 9 Ω and a capacitor 5.0 μF.



How much is the current I in the circuit in steady state ?

(A) 1.6 A

(B) 0.67 A

(C) 2.5 A

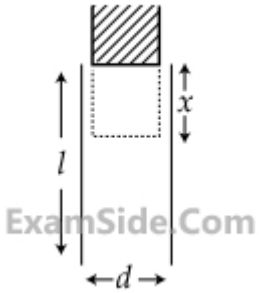
(D) 0.25 A

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Q161. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

A parallel plate capacitor is made of two plates of length l, width w and separated by distance d. A dielectric slab (dielectric constant K) that fits exactly between the plates is held near the edge of the plates. It is pulled into the capacitor by a force F = $-\frac{\partial U}{\partial x}$ where U is the energy of the capacitor when dielectric is inside the capacitor up to distance x (see figure). If the charge on the capacitor is Q then the force on the dielectric when it is near the edge is :



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(A) $\frac{Q^2 d}{2wl^2\in_0}K$
(B) $\frac{Q^2 w}{2dl^2\in_0}$ (K − 1)
(C) $\frac{Q^2 d}{2wl^2\in_0}$ (K − 1)
(D) $\frac{Q^2 w}{2dl^2\in_0}$ K

Q162. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

Three capacitances, each of 3 μF, are provided. These cannot be combined to provide the resultant capacitance of :

(A) 1 μF
(B) 2 μF
(C) 4.5 μF
(D) 6 μF

Q163. JEE Main 2014 (Offline)

Open solution

A parallel plate capacitor is made of two circular plates separated by a distance $5\ mm$ and with a dielectric of dielectric constant 2.2 between them. When the electric field in the dielectric is $3 \times 10^4\ V/m$ the charge density of the positive plate will be close to:

(A) $6 \times 10^{-7}\ C/m^2$
(B) $3 \times 10^{-7}\ C/m^2$
(C) $3 \times 10^4\ C/m^2$
(D) $6 \times 10^4\ C/m^2$

Q164. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

A parallel plate capacitor having a separation between the plates d, plate area A and material with dielectric constant K has capacitance C_0. Now one-third of the material is replaced by another material with dielectric constant 2K, so that effectively there are two capacitors one with area $\frac{1}{3}$ A, dielectric constant 2K and another with area $\frac{2}{3}$ A and dielectric constant K. If the capacitance of this new capacitor is C then $\frac{C}{C_0}$ is

(A) 1
(B) $\frac{4}{3}$
(C) $\frac{2}{3}$
(D) $\frac{1}{3}$

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Q165. JEE Main 2013 (Online) 22th April Morning Slot

To establish an instantaneous current of 2 A through a 1μF capacitor; the potential difference across the capacitor plates should be changed at the rate of:

(A) 2×10^4 V/s (B) 4×10^6 V/s (C) 2×10^6 V/s (D) 4×10^4 V/s

Open solution

Q166. JEE Main 2013 (Offline)

Two capacitors C_1 and C_2 are charged to $120\,V$ and $200\,V$ respectively. It is found that connecting them together the potential on each one can be made zero. Then

(A) $5C_1 = 3C_2$ (B) $3C_1 = 5C_2$ (C) $3C_1 + 5C_2 = 0$ (D) $9C_1 = 4C_2$

Open solution

Q167. AIEEE 2012 (Online) 26th May

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: It is not possible to make a sphere of capacity 1 farad using a conducting material.
Statement 2: It is possible for earth as its radius is 6.4×10^6 m.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

(B) Statement 1 is false, Statement 2 is true.

(C) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

(D) Statement 1 is true, Statement 2 is false.

Open solution

Q168. AIEEE 2012 (Online) 26th May

The capacitor of an oscillatory circuit is enclosed in a container. When the container is evacuated, the resonance frequency of the circuit is $10\,\text{kHz}$. When the container is filled with a gas, the resonance frequency changes by $50\,\text{Hz}$. The dielectric constant of the gas is

(A) 1.001 (B) 2.001 (C) 1.01 (D) 3.01

Open solution

Q169. AIEEE 2012 (Online) 12th May

A series combination of n_1 capacitors, each of capacity C_1 is charged by source of potential difference $4V$. When another parallel combination of n_2 capacitors each of capacity C_2 is charged by a source of potential difference V, it has the same total energy stored in it as the first combination has. The value of C_2 in terms of C_1 is then

(A) $16\frac{n_2}{n_1}C_1$ (B) $\frac{2C_1}{n_1 n_2}$ (C) $2\frac{n_2}{n_1}C_1$ (D) $\frac{16C_1}{n_1 n_2}$

Open solution

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Q170. AIEEE 2012 (Online) 12th May

A resistance R and a capacitance C are connected in series to a battery of negligible internal resistance through a key. The key is closed at $t = 0$. If after t sec the voltage across the capacitance was seven times the voltage across R, the value of t is

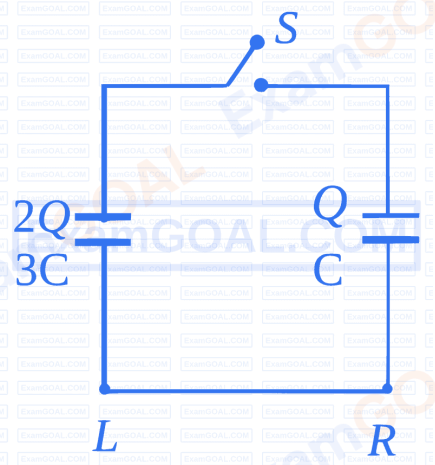
Open solution

(A) $3RC \ln 2$ (B) $2RC \ln 2$ (C) $2RC \ln 7$ (D) $3RC \ln 7$

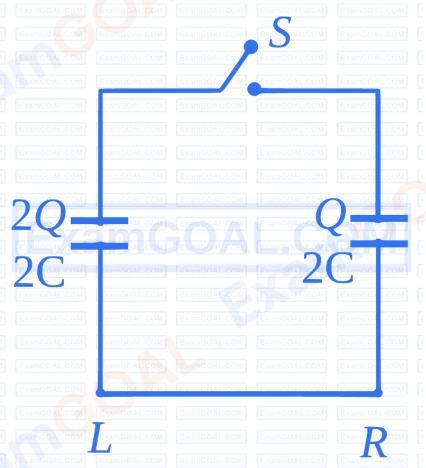
Q171. AIEEE 2012 (Online) 7th May

Two circuits (a) and (b) have charged capacitors of capacitance C, $2C$ and $3C$ with open switches. Charges on each of the capacitor are as shown in the figures. On closing the switches

Open solution



Circuit (a)



Circuit (b)

(A) No charge flows in (a) but charge flows from R to L in (b)

(B) Charges flow from L to R in both (a) and (b)

(C) Charges flow from R to L in (a) and from L to R in (b)

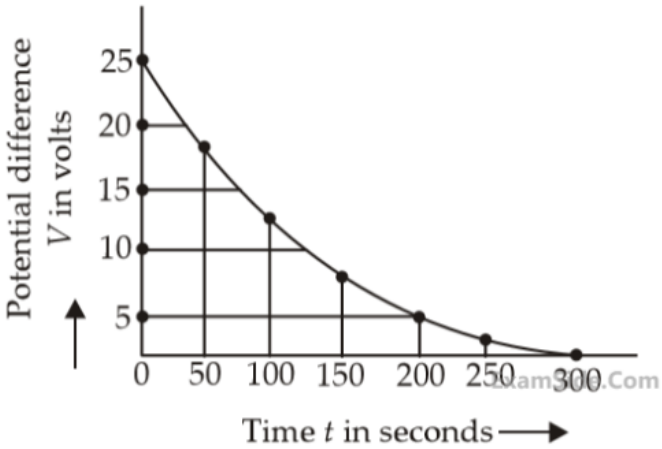
(D) No charge flows in (a) but charge flows from L to R in (b)

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Q172. AIEEE 2012

The figure shows an experimental plot for discharging of a capacitor in an R-C circuit. The time constant τ of this circuit lies between

Open solution



(A) 100 sec and 150 sec
(B) 0 and 50 sec
(C) 50 sec and 100 sec
(D) 150 sec and 200 sec

Q173. AIEEE 2011

A resistor 'R' and $2\mu F$ capacitor in series is connected through a switch to $200\ V$ direct supply. Across the capacitor is a neon bulb that lights up at $120\ V$. Calculate the value of R to make the bulb light up $5\ s$ after the switch has been closed. $(\log_{10}2.5 = 0.4)$

Open solution

(A) $1.7 \times 10^5\ \Omega$
(B) $2.7 \times 10^6\ \Omega$
(C) $3.3 \times 10^7\ \Omega$
(D) $1.3 \times 10^4\ \Omega$

Q174. AIEEE 2010

Let C be the capacitance of a capacitor discharging through a resistor R. Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio t_1/t_2 will be

Open solution

(A) 1
(B) $\frac{1}{2}$
(C) $\frac{1}{4}$
(D) 2

Q175. AIEEE 2008

A parallel plate capacitor with air between the plates has capacitance of $9\ pF$. The separation between its plates is 'd'. The space between the plates is now filled with two dielectrics. One of the dielectrics has dielectric constant $k_1 = 3$ and thickness $\frac{d}{3}$ while the other one has dielectric constant $k_2 = 6$ and thickness $\frac{2d}{3}$. Capacitance of the capacitor is now

Open solution

(A) $1.8\ pF$
(B) $45\ pF$
(C) $40.5\ pF$
(D) $20.25\ pF$

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Q176. AIEEE 2007

A parallel plate condenser with a dielectric of dielectric constant K between the plates has a capacity C and is charged to a potential V volt. The dielectric slab is slowly removed from between the plates and then reinserted. The net work done by the system in this process is

Open solution

(A) zero
(B) $\frac{1}{2}(K-1)CV^2$
(C) $\frac{CV^2(K-1)}{K}$
(D) $(K-1)CV^2$

Q177. AIEEE 2007

A battery is used to charge a parallel plate capacitor till the potential difference between the plates becomes equal to the electromotive force of the battery. The ratio of the energy stored in the capacitor and the work done by the battery will be

Open solution

(A) $1/2$
(B) 1
(C) 2
(D) $1/4$

Q178. AIEEE 2005

A parallel plate capacitor is made by stacking n equally spaced plates connected alternatively. If the capacitance between any two adjacent plates is $'C'$ then the resultant capacitance is

Open solution

(A) $(n+1)C$
(B) $(n-1)C$
(C) nC
(D) C

Q179. AIEEE 2005

A fully charged capacitor has a capacitance $'C'$. It is discharged through a small coil of resistance wire embedded in a thermally insulated block of specific heat capacity $'s'$ and mass $'m'$. If the temperature of the block is raised by $'\Delta T'$, the potential difference $'v'$ across the capacitor is

Open solution

(A) $\frac{mC\Delta T}{s}$
(B) $\sqrt{\frac{2mC\Delta T}{s}}$
(C) $\sqrt{\frac{2ms\Delta T}{C}}$
(D) $\frac{ms\Delta T}{C}$

Q180. AIEEE 2003

A sheet of aluminium foil of negligible thickness is introduced between the plates of a capacitor. The capacitance of the capacitor

Open solution

(A) decreases
(B) remains unchanged
(C) becomes infinite
(D) increases

Q181. AIEEE 2003

The work done in placing a charge of 8×10^{-18} coulomb on a condenser of capacity 100 micro-farad is

Open solution

(A) 16×10^{-32} $joule$
(B) 3.1×10^{-26} $joule$
(C) 4×10^{-10} $joule$
(D) 32×10^{-32} $joule$

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Q182. AIEEE 2002

If there are n capacitors in parallel connected to V volt source, then the energy stored is equal to

(A) CV (B) $\frac{1}{2}nCV^2$ (C) CV^2 (D) $\frac{1}{2n}CV^2$

Open solution

Q183. AIEEE 2002

Capacitance (in F) of a spherical conductor with radius $1\ m$ is

(A) 1.1×10^{-10} (B) 10^{-6} (C) 9×10^{-9} (D) 10^{-3}

Open solution

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Answer Key

Numerical

1. **2**	2. **200**	3. **8**	4. **4**	5. **15**
6. **5**	7. **64**	8. **1320**	9. **16**	10. **80**
11. **4**	12. **86**	13. **750**	14. **60**	15. **2**
16. **2**	17. **36**	18. **81**	19. **400**	20. **5**
21. **75**	22. **4**	23. **6**	24. **5**	25. **3**
26. **1**	27. **55**	28. **225**	29. **6**	30. **105**
31. **3**	32. **10**	33. **240**	34. **60**	35. **6**
36. **100**	37. **23**	38. **125**	39. **6**	40. **4**
41. **100**	42. **2**	43. **161**	44. **16**	45. **864**
46. **2**	47. **3**	48. **8**	49. **4**	50. **6**

MCQ (Single Correct Answer)

51. **B**	52. **A**	53. **A**	54. **A**	55. **B**
56. **A**	57. **C**	58. **D**	59. **D**	60. **A**
61. **D**	62. **C**	63. **A**	64. **A**	65. **A**
66. **D**	67. **D**	68. **D**	69. **B**	70. **B**
71. **D**	72. **A**	73. **D**	74. **A**	75. **B**
76. **B**	77. **A**	78. **B**	79. **D**	80. **D**
81. **C**	82. **D**	83. **C**	84. **A**	85. **A**
86. **C**	87. **A**	88. **A**	89. **A**	90. **A**
91. **D**	92. **A**	93. **A**	94. **A**	95. **C**
96. **A**	97. **A**	98. **A**	99. **D**	100. **B**
101. **C**	102. **A**	103. **A**	104. **C**	105. **C**
106. **C**	107. **A**	108. **C**	109. **B**	110. **B**
111. **C**	112. **B**	113. **A**	114. **D**	115. **B**
116. **D**	117. **D**	118. **B**	119. **D**	120. **C**
121. **B**	122. **C**	123. **C**	124. **B**	125. **D**
126. **D**	127. **C**	128. **A**	129. **C**	130. **A**
131. **C**	132. **A**	133. **A**	134. **B**	135. **B**
136. **A**	137. **D**	138. **A**	139. **B**	140. **A**
141. **A**	142. **A**	143. **B**	144. **C**	145. **B**

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146. **A**	147. **A**	148. **A**	149. **C**	150. **D**
151. **C**	152. **D**	153. **B**	154. **B**	155. **A**
156. **A**	157. **D**	158. **D**	159. **C**	160. **B**
161. **C**	162. **D**	163. **A**	164. **B**	165. **C**
166. **B**	167. **D**	168. **C**	169. **D**	170. **A**
171. **C**	172. **A**	173. **B**	174. **C**	175. **C**
176. **A**	177. **A**	178. **B**	179. **C**	180. **B**
181. **D**	182. **B**	183. **A**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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